Dell Unisphere for PowerMax 10.3.0 Product Guide

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Overview of Unisphere for PowerMax

Unisphere for PowerMax is a web-based application that enables users to configure and manage PowerMax, VMAX All Flash, and VMAX storage systems.

For embedded deployments, Unisphere for PowerMax runs embedded on PowerMax 2500 or PowerMax 8500 only. External deployment is required for storage systems running code levels below PowerMaxOS 10 (6079).

The side panel has the following items when the **Overview** view is selected:

- **Overview**—View overview of the status of all storage systems that the Unisphere for PowerMax manages.
- **Planning**—Unisphere for PowerMax provides a planning dashboard that allows users to see historical and predicted headroom information. This information can be used to create and manage planning jobs.
- **Systems**—View home dashboard view for all storage systems
- **Performance**—Monitors and manages storage system performance data (Dashboards, Charts, Analyze, and Reports). See Understanding Performance Management for more information.
- **Configuration**—Allows the viewing and configuration of system profiles and provisioning templates.
- **VMware**—Views all the relevant storage-related objects at an ESXi server and helps the troubleshooting of storage performance-related issues at the ESXi server. See Understanding Unisphere support for VMware for more information.
- **Events**—Includes Alerts, Job List, and CloudIQ Feed.

  NOTE: For additional information about event and alerts, see the *Events and Alerts for PowerMax and VMAX Users Guide*.

- **Support**—Displays support information.

The side panel has the following items when the storage system-specific view is selected:

- **Overview**—View overview of the status of all storage systems that the Unisphere for PowerMax manages.
- **Dashboard**—View the following dashboards for a selected storage system: System Health, Storage Group (SG) compliance, Capacity, Performance, and Protection
- **Storage**—Manage storage (storage groups, service levels, storage resource pools, volumes, external storage, PowerMax File, and vVol dashboard). See Storage Management for more information.
- **Hosts**—Manage hosts (hosts, masking views, port groups, initiators, PowerPath, mainframe, and CU images). See Host Management for more information.
- **Data Protection**—Manage data protection (snapshot policies, MetroDR, SRDF groups, migrations, virtual witness, File Protection, File Migration, open replicator, and device groups). See Data protection management for more information.
- **Performance**—Monitor and manage storage system performance data (Dashboards, Charts, Analyze, Heatmap, Reports, and Plan). See Understanding Performance Management for more information.
- **System**—Includes Hardware, System Properties, Physical View, File (eNAS), iSCSI, iSCSI+NVMe, and File configuration.
- **Events**—Includes Alerts, CloudIQ Feed, Job List, and Audit log.

  NOTE: For additional information about event and alerts, see the *Events and Alerts for PowerMax and VMAX Users Guide*.

- **Support**—Displays support information.

The following options are available from the title bar:

- View and configure settings.
- View newly added features.
- Discover systems
- Refresh system information.
- Search for objects
- View and manage jobs.
- View and manage alerts.
- View and manage settings.
- View profile
- Sign out
- View online help

A Unisphere for PowerMax Representational State Transfer (REST) API is also available. The API enables you to access diagnostic, performance and configuration data, and also enables you to perform provisioning operations on the storage system.

Information about the installation of Unisphere for PowerMax can be found in the *Unisphere for PowerMax Installation Guide* at the Dell support website.

For information specific to this Unisphere product release, see the *Unisphere for PowerMax Release Notes* at the Dell support website.

Your suggestions help continue to improve the accuracy, organization, and overall quality of the user publications. Send your feedback to content feedback.

Functionality supported by each operating system type

Unisphere enables the user to configure and manage PowerMax, VMAX All Flash, and VMAX storage systems.

Unisphere is a HTML5 web-based application that allows you to configure and manage PowerMax, VMAX All Flash, and VMAX storage systems.

NOTE: The following tables outline the functionality supported by each operating system type. In addition, depending on the deployment choice (host or embedded) and configuration, some of the functionality may not be supported.

Table 1. Functionality supported at storage system level depending on operating system type

Functionality	HYPERMAX OS 5977	PowerMaxOS 5978	PowerMaxOS 10 (6079)
Overview	✓	✓	✓
Dashboard	✓	✓	✓
Storage > Storage Groups	✓	✓	✓
Storage > Service Levels	✓	✓	✓
Storage > Storage Resource Pools	✓	✓	✓
Storage > Volumes	✓	✓	✓
Storage > External Storage	✓	✓	✓
Storage > vVol Dashboard	✓	✓	✓
Storage > File	N/A	N/A	✓
Hosts > Hosts	✓	✓	✓
Hosts > Masking Views	✓	✓	✓
Hosts > Port Groups	✓	✓	✓
Hosts > Initiators	✓	✓	✓
Hosts > PowerPath	N/A	✓	✓

Table 1. Functionality supported at storage system level depending on operating system type (continued)

Functionality	HYPERMAX OS 5977	PowerMaxOS 5978	PowerMaxOS 10 (6079)
Hosts > Mainframe	✓	✓	✓
Data Protection > Manage SRDF SGs	✓	✓	✓
Data Protection > Snapshot Policies	N/A	✓ - applies to storage systems running PowerMaxOS 5978 Q3 2020 and later.	✓
Data Protection > MetroDR	N/A	✓ - applies to storage systems running PowerMaxOS 5978 Q3 2020 and later.	✓
Data Protection > SRDF Groups	✓	✓	✓
Data Protection > Migrations	✓	✓	✓
Data Protection > Virtual Witness	✓	✓	✓
Data Protection > File Protection	N/A	N/A	✓
Data Protection > File Migration	N/A	N/A	✓
Data Protection > Open Replicator	✓	✓	✓
Data Protection > Device Groups	✓	✓	✓
Performance > Dashboards	✓	✓	✓
Performance > Charts	✓	✓	✓
Performance > Analyze	✓	✓	✓
Performance > Heatmap	✓	✓	✓
Performance > Reports	✓	✓	✓
Performance > Plan	✓	✓	✓
System > Hardware	✓	✓	✓

Table 1. Functionality supported at storage system level depending on operating system type (continued)

Functionality	HYPERMAX OS 5977	PowerMaxOS 5978	PowerMaxOS 10 (6079)
System > System Properties	✓	✓	✓
System > Physical View	N/A	N/A	✓ - applies to storage systems running PowerMaxOS 10.1.0 and later.
System > iSCSI	✓	✓	N/A
System > iSCSI + NVMe	N/A	N/A	✓
System > File (eNAS)	✓	✓	N/A
System > File Configuration	N/A	N/A	✓
Events > Alerts	✓	✓	✓
Events > CloudIQ Feed	✓	✓	✓
Events > Jobs	✓	✓	✓
Events > Audit Log	✓	✓	✓
Serviceability > Serviceability	N/A	N/A	✓
Serviceability > Applications	N/A	N/A	✓
Serviceability > Updates	N/A	N/A	✓

Table 2. Functionality supported at overview level and systems level depending on operating system type

Functionality	HYPERMAX OS 5977	PowerMaxOS 5978	PowerMaxOS 10 (6079)
Overview	✓	✓	✓
Planning	✓	✓	✓
Systems	✓	✓	✓
Performance > Dashboards	✓	✓	✓
Performance > Charts	✓	✓	✓

Table 2. Functionality supported at overview level and systems level depending on operating system type (continued)

Functionality	HYPERMAX OS 5977	PowerMaxOS 5978	PowerMaxOS 10 (6079)
Performance > Analyze	✓	✓	✓
Performance > Reports	✓	✓	✓
Configuration > System Profiles	N/A	N/A	✓ (applies to local storage systems running PowerMaxOS 10 (6079).)
Configuration > Provisioning Templates	✓	✓	✓
VMware > vCenters and ESXi	✓	✓	✓
Events > Alerts	✓	✓	✓
Events > CloudIQ Feed	✓	✓	✓
Events > Jobs	✓	✓	✓
Support	✓	✓	✓

Capacity information

Unisphere for PowerMax supports measurement of capacity using both the base 2 (binary) and base 10 (decimal) systems.

Storage capacity can be measured using two different systems – base 2 (binary) and base 10 (decimal). Organizations such as the International System of Units (SI) recommend using the base 10 measurement to describe storage capacity. In base 10 notation, one MB is equal to 1 million bytes, and one GB is equal to 1 billion bytes.

Operating systems generally measure storage capacity using the base 2 measurement system. Unisphere for PowerMax and Solutions Enabler use the base 2 measurement system to display storage capacity with the TB notation as it is more universally understood. In base 2 notation, one MB is equal to 1,048,576 bytes and one GB is equal to 1,073,741,824 bytes.

Table 3. Capacity information

Name	Abbreviation	Binary Power	Binary Value (in Decimal)	Decimal Power	Decimal (Equivalent)
kilobyte	KB	2^{10}	1,024	10^{3}	1,000
megabyte	MB	2^{20}	1,048,576	10^{6}	1,000,000
GB	GB	2^{30}	1,073,741,824	10^{9}	1,000,000,000
terabyte	TB	2^{40}	1,099,511,627,776	10^{12}	1,000,000,000,000

Understanding resource usage

Unisphere supports a view of capacity utilization on a storage system throughout its life cycle.

The Unisphere capacity dashboard supports you to understand the storage system resources being used in the following scenarios:

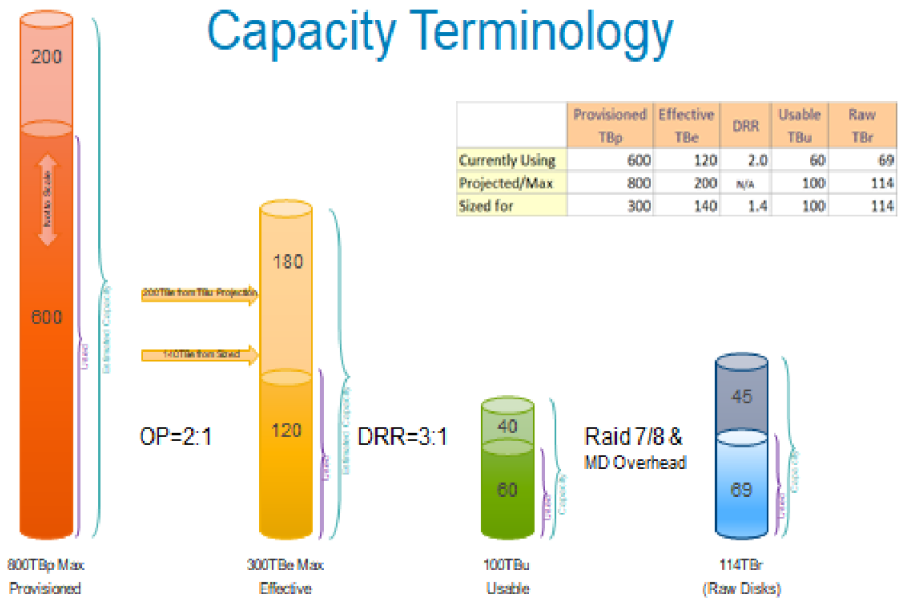
Day 1: The storage system is installed and is ready for production:

- Can I see the physical disk capacity of the system?
- If I have purchased different disk technology tiers, can I see the capacity of each tier?
- Can I see how much effective capacity the system was sized for?

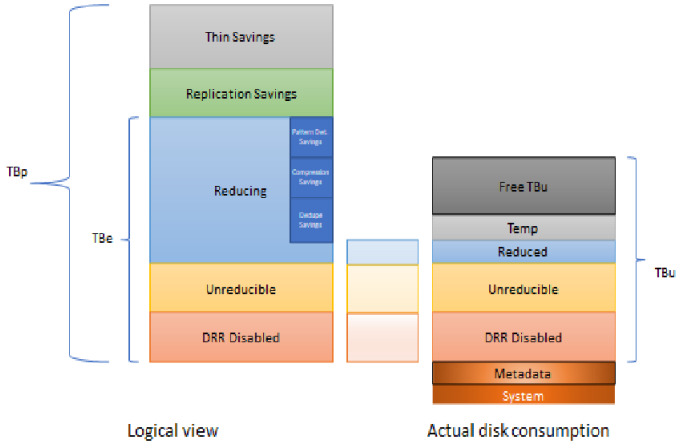
Day N: The storage system is in production, and storage is being provisioned:

- How much more can I write?
- How much more effective capacity is available?
- How much physical capacity is consumed?
- How much storage is allocated?
- How much of the system is over provisioned?
- What is the overall system Data Reduction Ratio (DRR)?
- How much capacity is not enabled for DRR?
- How much capacity is not reducible?
- How much capacity is being used by system resources that cannot be used for storage provisioning?
- How much system capacity is consumed by local replication?
- How much capacity is available for snapshots?

An overview of capacity terminology is displayed in the following figure.



A logical view in comparison with a disk consumption view is displayed in the following figure.



Login authentication

Unisphere authenticates users attempting to access the system.

See Configure authentication authorities for information about configuring authentication authorities.

When you log in, Unisphere checks the following authorities:

- **Windows**—The user has a Windows account on the server. (Log in to Unisphere with your Windows Domain\Username and Password.)
- **LDAP-SSL**—The user account is stored on an LDAP-SSL server (Log in to with your Unisphere LDAP-SSL Username and Password). The Unisphere Administrator or SecurityAdmin must set the LDAP-SSL server location in the LDAP-SSL Configuration dialog box.
- **Local**—The user has a local Unisphere account. Local user accounts are stored locally on the Unisphere server host. (Log in to Unisphere with your Username and Password.)
- **RSA SecurID MFA**—If RSA SecurID MFA authentication is enabled, the RSA token must be entered into the password field immediately followed by the user's password.
- **YubiKey MFA**—If MFA YubiKey authentication is enabled, the one-time password(OTP) (press button on the YubiKey) is entered into the password field immediately followed by the user entering the user password.

User names are case-sensitive and allow alphanumeric characters of either case, an underscore, a dash, or a period:

- a-z
- A-Z
- 0-9
- _
- .
- -

There are no restrictions on special characters when using passwords.

The Initial Setup User (ISU) is used to create the user account or add authorization rules to existing LDAP/AD or Windows operating system users.

View and manage YubiKey Device ID mapping

Unisphere for PowerMax supports MFA YubiKey. YubiKey is a hardware device that provides authentication/One Time Password (OTP) for multi-factor authentication. YubiKey is a small USB security key that protects access to configured devices.

Prerequisites

You must be a Security Admin or Admin to perform this task.

You must have a YubiKey fob.

You have created a local user that plans to use YubiKey.

About this task

NOTE: YubiKey MFA and RSA MFA are mutually exclusive.

Steps

1. Select  .
2. Select **Security** > **YubiKey Device ID Mapping**.

   The following properties display:

   - **Username**—Username
   - **YubiKey Device ID**—YubiKey device identifier

   The following controls are available:

   - **Attach Device ID**—allows you to add a username and attach one or more device IDs. The device ID can be typed in, copied and pasted from the YubiKey fob, or entered by clicking the fob.
   - **Modify Device ID**—allows you to modify the attached device IDs.
   - **Remove Device ID**—allows you to remove device IDs.

Next steps

Create Yubico API Key and enable YubiKey (see Configure authentication authorities).

Configure authentication authorities

Users gain access to a storage system or component either directly through a role assignment or indirectly through membership in a user group that has a role assignment, or both.

Prerequisites

When configuring authentication to use LDAP, obtain the LDAP-SSL server bind distinguished name (DN) and password from your LDAP Administrator.

When configuring authentication to use YublKey, create the Yubico API key first. Go to Get API, enter your email address and YubiKey OTP, and click **Get API key**. Copy Client ID and Secret Key to use in step 7.

About this task

This procedure explains how to configure Unisphere to authenticate users.

NOTE: YubiKey and RSA MFA are mutually exclusive.

To configure authentication:

Steps

1. Select  to open the **Settings** panel.
2. Select **Users and Groups** > **Authentication**.
3. Select the **Authentication Authority** to use during login. Possible values are:
   - **Local Directory**—You can disable this authority when it is enabled, and enable it when it is disabled. When it is enabled, users can log in as a user from the CST local directory.
   - **LDAP-SSL**—You can disable this authority when it is enabled, and enable it when it is disabled. When it is enabled, users can log in as a user from the configured LDAP directory.
   - **Windows OS/AD**—You can disable this authority when it is enabled, and enable it when it is disabled. When it is enabled, users can log in as a user from the Windows local host or from the Active Directory domain, or both. This option applies only to Windows installations.
   - **RSA SecurID**—When it is enabled, the RSA token must be entered into the password field immediately followed by the user password.
   - **YubiKey**—When YubiKey is enabled, the OTP (press button on the YubiKey) is entered into the password field immediately followed by the user entering the user password.

- **OIDC SSO**—When OIDC SSO is enabled, you can log in using OIDC SSO.

4. If you select the **Windows OS/AD** authority and click **Modify**, as an option you can specify to limit authentication to members of a specific Windows **OS/AD** group. To limit authentication, select the **Limit authentication to members of a specific Windows OS/AD group(s)** checkbox option, and type the **Group Name(s)**, separated by commas.
5. If you are configuring LDAP-SSL, click **Enable** or **Modify** and do the following:
   a. Specify values for the following parameters.
      - **Server (IP or Hostname)**—IP address or hostname of the LDAP server to use for authentication - only alphanumeric characters are allowed.
      - **Port**—Port number of the LDAP service. Typically, this value is 636 or 3269 for LDAP-SSL. Valid values range from 1 through 65,535.
      - **Bind DN**—Distinguished name of the privileged account used to perform operations, such as searching users and groups, on the LDAP directory - only alphanumeric characters are allowed.
      - **Bind Password**—Password of the privileged account- only alphanumeric characters are allowed.
      - **User Search Path**—Distinguished name of the node at which to begin user searches. Only alphanumeric characters are allowed.
      - **User Object Class**—Object class identifying users in the LDAP hierarchy - only alphanumeric characters are allowed.
      - **User ID Attribute**—Attribute identifying the user login ID within the user object- only alphanumeric characters are allowed.
      - **Group Search Path**—Distinguished name of the node at which to begin group searches - only alphanumeric characters are allowed.
      - **Group Object Class**—Object class identifying groups in the LDAP hierarchy - only alphanumeric characters are allowed.
      - **Group Name Attribute**—Attribute identifying the group name - only alphanumeric characters are allowed.
      - **Group Member Attribute**—Attribute indicating group membership for a user within the group object - only alphanumeric characters are allowed.
   b. To upload an SSL certificate, click **Choose File**, locate the certificate, and click **Open**. To view the contents of the certificate, click **View Certificate**. To clear the file selection, click **Clear**.
   c. Optional: To limit authentication to only members of specific LDAP groups, click **Limit Authentication to members of LDAP group(s)**, select the option, and then type the **Group Name(s)**, separated by commas.
6. If you are configuring RSA SecurID, click **Enable** or **Modify** and do the following:

   Specify values for the following parameters:
   - **Server (IP or Hostname)**—IP address or hostname of the RSA SecurID server to use for authentication. Only alphanumeric characters are allowed.
   - **Port**—Port number
   - **Client ID**—Client identity
   - **Token Length**—Token length
   - **Access key**—Access key value - click **Show** to display the access key value and click **Hide** to hide the access key value.
   - To upload a root CA certificate of the RSA SecurID server click **Choose File**, locate the certificate, and click **Open**. To clear the file selection, click **Clear**.
   - **RSA Username**—RSA username
   - **RSA Token**—RSA token - click the eye icon when typing in the value for the token and you can see what you are entering.
   - **REST API user exception**—Select to configure a REST API user exception.
   - **Username(s)**—List the comma-separated usernames to which the exception applies.
7. If you are configuring YubiKey, click **Enable** or **Modify** and do the following:

   Specify values for the following parameters:
   - **Server (IP or Hostname)**—IP address or hostname of the YubiKey server to use for authentication. Only alphanumeric characters are allowed.
   - **Port**—Port number
   - **Client ID**—Client identity (see Pre-requisites)
   - **API key**—API key value (see Pre-requisites) - click **Show** to display the access key value and click **Hide** to hide the access key value.
   - To upload a root CA certificate of the YubiKey server, click **Choose File**, locate the certificate, and click **Open**. To clear the file selection, click **Clear**.
   - **YubiKey Username**—YubiKey username
   - **YubiKey OTP**—YubiKey token- click the eye icon when typing in the value and you can see what you are entering.
   - **YubiKey REST API user exception**—Select to configure a REST API user exception.

- **Username(s)**—List the comma-separated usernames to which the exception applies.

8. If you are configuring OIDC SSO, click **Enable** or **Modify** and do the following:

   Specify values for the following parameters:
   - **Server (IP or Hostname)**—IP address or hostname of the OIDC SSO server to use for authentication. Only alphanumeric characters are allowed. Microsoft Entra ID is seleted by default.
   - **Port**—Port number
   - **OIDC Client ID**—Client identity
   - **OIDC Client Secret**—Client password
   - **OIDC Metadata URL** —Client password
   - To upload a root CA certificate of the OIDC SSO server, click **Choose File**, locate the certificate, and click **Open**. To clear the file selection, click **Clear**.

9. Click **OK**.

Understanding the system health score

The **Overview** dashboard provides a single place from which you can quickly determine the health of the system.

The system health score can help you spot where your most severe health issues are, based on the following core factors: Configuration, Capacity, Data Protection, and Performance. The area with the highest risk to your system health lowers the score until remedial actions are taken.

The **Overview** panel displays values for the following high-level health metrics: Configuration, Capacity, Data Protection, and Performance. It also displays an overall health score that is based on the lowest health score out of these metrics. The health score is calculated every five minutes. The overall value is always calculated from all metric values. If a health score category is seen as stale or unknown, the overall health score is not updated. The previously calculated overall health score is displayed but its value is denoted as stale by setting the menu item to gray.

The Capacity health score is based on % Effective Used Capacity. Capacity levels are checked at the SRP level and SRP Emulation level (where a mixed SRP emulation is involved).

The capacity health scores are calculated as follows:

- Fatal level - based on what is defined in the **System Threshold and Alerts** dialog. The default fatal threshold is 100% - 30 points.
- Critical level - based on what is defined in the **System Threshold and Alerts** dialog. The default critical threshold is 80% - 20 points.

The Capacity health score is also impacted by the CloudIQ Capacity Prediction analytics algorithm, which forecasts when the SRP is predicted to be full, based on historical capacity usage. CloudIQ sends a health score deduction to Unisphere for PowerMax if the predicted range is within a quarter or less. The extent of the deduction depends on how close it is to reaching full capacity.

The Performance health score is calculated using the threshold limits of the following categories and metrics:

- FE Director : % busy, queue depth utilization (queue depth utilization is not checked for EF directors).
- FE port: % busy
- BE port: % busy (not applicable for storage systems running PowerMaxOS 10 (6079))
- BE Director (DA): % busy
- SRDF port: % busy
- SRDF Director: % busy
- DX port: % busy (not applicable for storage systems running PowerMaxOS 10 (6079))
- External Director: % busy (not applicable for storage systems running PowerMaxOS 10 (6079))
- EDS Director: % busy (not applicable for storage systems running PowerMaxOS 10 (6079))
- Cache Partition: %WP utilization (not applicable for storage systems running PowerMaxOS 10 (6079))
- EM Director : %WP utilization (applicable for storage systems running PowerMaxOS 10 (6079))
- System (Array): % Cache WP Utilization

For each instance and metric for a particular category, the threshold info is found. If not set, use the default thresholds. The default thresholds are:

FE Port: % busy - Critical 70

FE Director: % busy - Critical 70 ; Queue Depth Utilization - Critical 75

BE Port: % busy - Critical 70

BE Director (DA): % busy - Critical 70

SRDF Port: % busy - Critical 70

SRDF Director: % busy - Critical 70

DX Port: - % busy - Critical 70

External Director: % busy- Critical 70

EDS Director: % busy - Critical 70

Cache Partition: %WP utilization - Critical 75

EM Director: % busy - Critical 70

System (Array): % Cache WP Utilization - Critical 60

The Performance health score also incorporates the following:

- Storage Group (SG)
  - Read Response Time: Critical - five points
  - Write Response Time: Critical - five points
  - Response Time: Critical - five points
- Service Level compliance
- Underperforming: - five points

For each storage group instance and metric for particular category, the threshold info is found. If not found, default thresholds are used.

The Response Time, Read Response Time, Write Response Time metrics are ignored from the health score calculation if the alert is not enabled for the SG/metric.

The Service Level Compliance health score is based on then Workload Planner (WLP) workload state. A reduction from the health score is performed when storage groups that have a defined service level are not meeting the service level requirements.

If the Service Level Compliance alert is disabled for the SG, it is ignored when performing the health score calculation.

The Performance score is calculated as follows:

- Critical level: - five points

The Data Protection health score is based on if there are SRDF Groups in an offline state or Transmit Idle State. Both these scenarios result in a -5 point deduction to the Data Protection Health Score.

SRDF Groups associated with local SRDF ports that are offline cause a reduction of -2.

The Configuration health score is calculated every five minutes and is based on the director and port alerts in the system at the time of calculation. Unisphere does not support alert correlation or auto clearing, you are required to manually delete alerts that have been dealt with or are no longer relevant as these alerts impact on the hardware health score until they are removed from Unisphere.

The Configuration health score is calculated as follows:

- Director out of service - 40 points
- Director Offline - 20 points
- Port Offline - 10 points

For embedded Unisphere systems that are running PowerMaxOS 6079 and that support PowerMax File, the configuration health score calculation is also impacted by the following PowerMax File alerts:

- Recovery failed on file system 'fsname' in NAS server 'nasserver' (fsid 'fsid') - 30 points
- File system 'fsname' in NAS server 'nasserver' (fsid 'fsid') is offline after discovering corruption - 30 points.
- File system 'fsname' in NAS server 'nasserver' (fsid 'fsid') is offline due to receiving an **I/O** error -30 points.
- Recovery failed on the 'fsname' file system in NAS server 'nasserver' (fsid 'fsid') - 30 points
- The 'fsname' file system in NAS server 'nasserver' (fsid 'fsid') is offline after discovering corruption. - 30 points
- The 'fsname' file system in NAS server 'nasserver' (fsid 'fsid') is offline due to receiving an I/O error - 30 points.
- NAS node 'node' is down - 20 points.
- NAS node 'node' is down and its automatic recovery has failed - 20 points.
- NAS server 'nasserver' is down - 10 points.
- NAS server 'nasserver' fault tolerance is degraded - 10 points.
- NAS server 'nasserver' is in maintenance mode - 10 points.
- The DNS client of the 'nasserver' NAS server is unable to connect to all configured DNS servers - 20 points.

- No LDAP servers that are configured for NAS server 'nasserver' are responding - 20 points.
- The LDAP service configuration of the NAS server 'nasserver' for domain 'domain' failed - 20 points
- The NIS client is unable to connect to all configured NIS servers - 20 points.
- The NAS server 'nasserver' in the domain 'domain' cannot reach any domain controller - 20 points.
- No virus checker server is available - 5 points.
- LDAP client settings on NAS server 'nasserver' are not valid within domain 'domain' - 5 points.
- The SMB server of the NAS server 'nasserver' is configured to be joined to the domain 'domain', but is currently not joined - 5 points.

For embedded Unisphere systems that are running PowerMaxOS 10 (6079) and that support **PowerMax File**, auto clearing of **PowerMax File** alerts is supported.

Auto clearing of alerts is a mechanism whereby if an alert of severity **Normal** arrives into the system, this alert clears the alerts relating to this normal alert that are already there and have a higher severity.

Manage jobs

Certain configuration tasks that are performed on a storage system may not be not immediately processed, but instead are kept in a job list for review and submission in batches.

About this task

One way to identify these tasks is from the dialog boxes. They have a button that is named **Add to Job List**.

Unisphere includes a job list view, from which you can view and manage the job list for a storage system.

Steps

1. Select the storage system.
2. Select **Events** > **Jobs**.

Unisphere for PowerMax alert monitoring recommendations

This topic outlines the list of recommended alerts for you to monitor or consider monitoring (depending on your environment) when configuring alert policies using Unisphere for PowerMax.

NOTE: These recommendations are for storage systems running HYPERMAX OS 5977, PowerMaxOS 5978, and PowerMaxOS 10 (6079).

Alert notifications should also be enabled for these alerts.

In addition, notifications should be configured for the default System Thresholds Alerts set and Notifications set.

It is recommended that you monitor the following:

- Array Component events
- Array Events
- Array - Deferred Service Threshold Alert
- Array - Director Status
- Array - Disk Status
- Array - Environmental Alert
- Array - Hotspare Invoked
- Array - Port Link status
- Array - Port status
- Array - RVA Spare Coverage
- Array - SP Alerts
- Array - SRDF Alerts
- Array - SRDF Job Flow Control Change
- Array - SRDF Link Status

- Array - SRDF/A No Cycle Switch Alert
- Array - SRDF/A Session
- Array - SRDF/A Session dropped, Transmit Idle state timeout
- Array - SRDF/A Session entering Transmit Idle state
- Array - SRDF/A Session recovered from a Transmit Idle state
- Array - Target Enginuity Warning

Consider monitoring the following depending on the customer environment:

- Array - Device Config Change
- Array - Device Status
- Array - Thin Device Allocation
- Array - Thin Device Usage

Alerts

You can configure Unisphere to monitor storage systems for specific events or error conditions. When an event or error of interest occurs, Unisphere for PowerMax displays an alert and, when configured to do so, notifies you of the alert by way of email, SNMP, or Syslog.

Server alerts

Server alerts are alerts that are generated by Unisphere for PowerMax itself.

Unisphere generates server alerts under the conditions that are listed in the following table:

Checks are run on 10 minute intervals, and alerts are raised on 24-hour intervals from the time the server was last started. These time intervals also apply to discover operations. That is, performing a discover operation does not force the delivery of these alerts.

NOTE: Runtime alerts are not storage system-specific. They can be deleted if the user has admin or storage admin rights on at least one storage system. A user with a monitor role cannot delete the server alerts.

Table 4. Server alerts

Server alert	Number of volumes	Threshold	Alert Details
Total memory on the Unisphere server	0–64,000	12 GB	System memory <number GB> is below the minimum requirement of <# GB>.
	64,000–128,000	16 GB	
	128,000–256,000	20 GB	
Free disk space on the Unisphere installed directory	0–64,000	100 GB	Free disk space <# GB> is below the minimum requirement of <# GB>
	64,000–128,000	140 GB	
	128,000–256,000	180 GB	
Number of managed storage systems	The threshold is 20.		The number of managed storage systems <#> is over the maximum supported number of #.
Number of managed volumes	256,000		The number of managed volumes <#> is over the maximum supported number of <#>. Solutions Enabler may indicate a slightly different number of volumes than indicated in this alert.
Number of gatekeepers	6		The number of gatekeepers <#> on the storage system is below the minimum requirement of 6.

Understanding licenses

Unisphere for powerMax supports electronic licensing (e-Licensing). e-Licensing is a license management solution to help you track and comply with software license entitlement.

e-Licensing uses embedded locking functions and back-office IT systems and processes. It provides better visibility into software assets, easier upgrade, and capacity planning. It reduces the risk of noncompliance, while still adhering to a strict "do no harm" policy to your operations.

When installing licenses with e-Licensing, you obtain license files from customer service, copy them to a Solutions Enabler or a Unisphere host, and load them onto storage systems.

Each license file fully defines the entitlements for a specific system, including its activation type (Individual or Enterprise), the licensed capacity, and the date the license was created. If you want to add a product title or increase the licensed capacity of an entitlement, obtain a new license file from online support and load it onto the storage system.

When managing licenses, Solutions Enabler, Unisphere for PowerMax, z/OS Storage Manager (EzSM), MF SCF native command line, TPF, and IBM i platform console, provide detailed usage reports that enable you to better manage capacity and compliance planning.

There are two types of e-Licenses: host-based and storage system-based. Host-based licenses, as the name implies, are installed on the host. And, storage system-based licenses are installed on the storage system. For information about the types of licenses and the features they activate, see the *Solutions Enabler Installation Guide*.

Unisphere enables you to add and view storage system-based licenses, and add, view, and remove host-based licenses.

Unisphere uses storage system-based e-Licensing.

For storage systems running PowerMaxOS 10 (6079), or later, only XML format licenses are supported. The installation of older format licenses for storage systems running PowerMax 5978, or earlier, is not supported.

NOTE:

For more information about e-Licensing, see the *Solutions Enabler Installation Guide*.

Understanding user authorization

User authorization is a tool for restricting the management operations users can perform on a storage system.

By default, user authorization is enabled for Unisphere users, regardless of whether it is enabled on the storage system.

When configuring user authorization, an Administrator or SecurityAdmin maps individual users or groups of users to specific roles on storage systems and these roles determine the operations that the users can perform. These user-to-role-to-storage system mappings (known as authorization rules) are maintained in the `symauth` users list file that is stored on the host or storage system, depending on the storage operating environment.

NOTE: If the `symauth` file contains one or more users, users who are not listed in the file are unable to access or even see storage systems from the Unisphere console.

Individual and group roles

Users gain access to a storage system or component either directly through a role assignment and/or indirectly through membership in a user group that has a role assignment.

If a user has two different role assignments (one as an individual and one as a member of a group), the permissions that are assigned to the user are combined. For example, if a user is assigned a Monitor role and a StorageAdmin role through a group, the user is granted Monitor and StorageAdmin rights.

Roles

A Unisphere user can assume several roles. Tasks and associated permissions are associated with each role.

The following lists the available roles.

You can assign up to four of these roles per authorization rule.

For a more detailed look at the permissions that go along with each role, see Roles and associated permissions.

- **None**—Provides no permissions.
- **Monitor**—Performs read-only (passive) operations on a storage system excluding the ability to read the audit log or access control definitions.
- **StorageAdmin**—Performs all management (active or control) operations on a storage system and modifies GNS group definitions in addition to all Monitor operations.
- **Admin**—Performs all operations on a storage system, including security operations, in addition to all StorageAdmin and Monitor operations.
- **SecurityAdmin**—Performs security operations on a storage system, in addition to all Monitor operations.
- **Auditor**—Grants the ability to view, but not modify, security settings for a storage system(including reading the audit log, **symacl** list, and **symauth**) in addition to all Monitor operations. This role is the minimum role that is required to view the storage system audit log.
- **Local Replication**—Performs local replication operations (SnapVX or legacy Snapshot, Clone, BCV). When creating Secure SnapVX snapshots a user must have Storage Admin rights at the storage system level. This role also automatically includes Monitor rights.
- **Remote Replication**—Performs remote replication (SRDF) operations involving devices and pairs. Users can create, operate on, or delete SRDF device pairs but cannot create, modify, or delete SRDF groups. This role also automatically includes Monitor rights.
- **Device Management**—Grants user rights to perform control and configuration operations on devices. Storage Admin rights are required to create, expand, or delete devices. This role also automatically includes Monitor rights.
- **PerfMonitor**—Performs read-only (passive) operations on performance management functionality.
- **MainframeAdmin**—Performs operations on mainframe objects, such as splits and CU images, and performs z/OS mapping and unmapping. This role has Monitor rights for non-mainframe operations.

In addition to these user roles, Unisphere includes an administrative role, the Initial Setup User. This user (defined during installation) is a temporary role that provides administrator-like permissions for adding local users and roles to Unisphere.

User IDs

Users and user groups are mapped to their respective roles by IDs.

These IDs consist of a three-part string in the form:

`Type:Domain\Name`

where:

- Type—Specifies the type of security authority that is used to authenticate the user or group. Possible types are:
  - L—Indicates a user or group that LDAP authenticates. In this case, Domain specifies the domain controller on the LDAP server. For example: `L:danube.com\Finance` indicates that user group Finance has logged in through the domain controller danube.com
  - C—Indicates a user or group that the Unisphere server authenticates. For example: `C:Boston\Legal` indicates that user group Legal logged in through Unisphere server Boston.
  - H—Indicates a user or group that is authenticated by logging in to a local account on a Windows host. In this case, Domain specifies the hostname. For example: `H:jupiter\mason` indicates that user mason logged in on host jupiter.
  - D—Indicates a user or group that is authenticated by a Windows domain. In this case, Domain specifies the domain or realm name. For example: `D:sales\putman` indicates that the user **`putman`** has logged in through a Windows domain sales.
- Name—specifies the username relative to that authority. It cannot be longer than 32 characters, and spaces are allowed when delimited with quotes. Usernames can be for individual users or user groups.

Within role definitions, IDs can be either fully qualified (as shown above), partially qualified, or unqualified. When the Domain portion of the ID string is an asterisk (*), the asterisk is treated as a wildcard, meaning any host or domain.

The Domain portion of the ID must be fully qualified when configuring group access.

For example:

- `D:ENG\jones`—Fully qualified path with a domain and username (for individual domain users)
- `D:ENG.xyz.com\ExampleGroup`—Fully qualified domain name and group name (for domain groups)
- `D:*\jones`—Partially qualified that matches username **jones** with any domain
- `H:HOST\jones`—Fully qualified path with a hostname and username
- `H:*\jones`—Partially qualified that matches the username **jones** within any host

- `jones`—Unqualified username that matches any **jones** in any domain on any host

If a user is matched by more than one mapping, the user authorization mechanism uses the more specific mapping. If an exact match (for example, `D:sales\putman`) is found, that is used. If a partial match (for example, `D:*\putman`) is found, that is used. If an unqualified match (for example, **putman**) is found, that is used. Otherwise, the user is assigned a role of None.

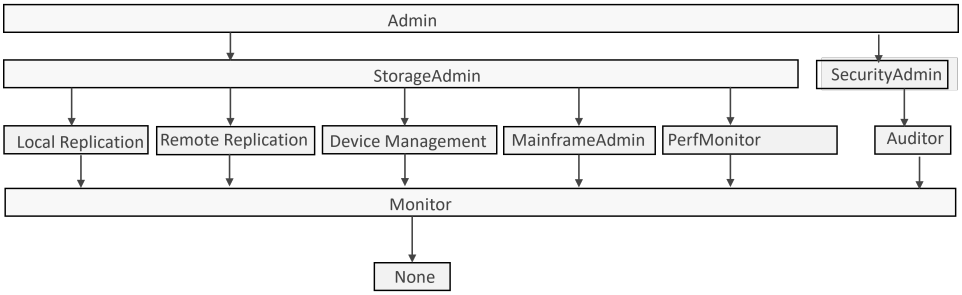
Roles and associated permissions

Users gain access to a storage system or component directly through a role assignment or indirectly through membership in a user group that has a role assignment.

A Unisphere user can assume several roles. Tasks and associated permissions are associated with each role. The Role Based Access Control (RBAC) feature provides a method for restricting the management operations that individual users or groups of users may perform on storage systems.

If a user has two different role assignments (one as an individual and one as a member of a group), the permissions that are assigned to the user are combined. For example, if a user is assigned a Monitor role and a StorageAdmin role through a group, the user is granted Monitor and StorageAdmin rights.

The following diagram outlines the role hierarchy.



Roles are assigned as part of the user creation process (see Create local users).

The following tables detail the permissions that are associated with each role in Unisphere. You can assign up to four of these roles per authorization rule.

NOTE: In addition to these user roles, Unisphere includes an administrative role, the **Initial Setup User**. This role (defined during installation) is a temporary role that provides administrator-like permissions for adding local users and roles to Unisphere.

The roles (and the acronyms that are used for the roles) in these tables are:

- **None**—Provides no permissions.
- **Monitor (MO)**—Performs read-only (passive) operations on a storage system excluding the ability to read the audit log or access control definitions.
- **StorageAdmin (SA)**—Performs all management (active or control) operations on a storage system and modifies GNS group definitions in addition to all Monitor operations.
- **Admin (AD)**—Performs all operations on a storage system, including security operations, in addition to all StorageAdmin and Monitor operations.
- **SecurityAdmin (SecA)**—Performs security operations on a storage system, in addition to all Monitor operations.
- **Auditor (AUD)**—Grants the ability to view, but not modify, security settings for a storage system (including reading the audit log, `symacly` list, and `symauth`) in addition to all Monitor operations. It is the minimum role that is required to view the storage system audit log.
- **PerfMonitor (PM)**—Performs read-only (passive) operations on performance management functionality. This role also includes Monitor rights.
- **MainframeAdmin (MA)**—Grants user rights to perform mainframe control and configuration operations on Splits and CU images. This role also grants the rights to perform z/OS Map and Unmap operations on CKD devices. This role also includes Monitor rights for non-mainframe operations.
- **Local Replication**—Performs local replication operations (SnapVX or legacy Snapshot, Clone, BCV). To create Secure SnapVX snapshots, a user must have Storage Admin rights at the storage system level. This role also includes Monitor rights.
- **Remote Replication**—Performs remote replication (SRDF) operations involving devices and pairs. Users can create, operate upon, or delete SRDF device pairs but cannot create, modify, or delete SRDF groups. This role also includes Monitor rights.
- **Device Management**—Grants user rights to perform control and configuration operations on devices. Storage Admin rights are required to create, expand, or delete devices. This role also includes Monitor rights.

NOTE: The RBAC roles for performing local and remote replication actions are outlined in Roles for performing local and remote replication actions.

NOTE: The RBAC roles for SRDF local and remote replication actions are outlined in RBAC roles for SRDF local and remote replication actions.

NOTE: The RBAC roles for TimeFinder SnapVX local and remote replication actions are outlined in RBAC roles for TimeFinder SnapVX local and remote replication actions.

Table 5. User roles and associated permissions

Permissions	AD	SA	MO	SecA	AUD	None	PM	MA
Create/delete user accounts	Yes	No	No	Yes	No	No	No	No
Reset user password	Yes	No	No	Yes	No	No	No	No
Create roles	Yes	Yes	No	Yes (self-excluded)	No	No	No	No
Change own password	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Change NVMe secure provisioning restrictions	Yes	No	No	Yes	No	No	No	No
Manage storage systems	Yes	Yes	No	No	No	No	No	No
Discover storage systems	Yes	No	No	Yes	No	No	No	No
Add/show license keys	Yes	Yes	No	No	No	No	No	No
Set alerts and Optimizer monitoring options	Yes	Yes	No	No	No	No	No	No
Release storage system locks	Yes	Yes	No	No	No	No	No	No
Set Access Controls	Yes	Yes	No	No	No	No	No	No
Set replication and reservation preferences	Yes	Yes	No	No	No	No	No	No
View and export the storage system audit log	Yes	No	No	Yes	Yes	No	No	No
Access performance data	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes
Start data traces	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes
Set performance thresholds/alerts	Yes	Yes	No	No	No	No	Yes	Yes
Create and manage performance dashboards	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes

Table 6. Permissions for Local Replication, Remote Replication, and Device Management roles

Permissions	Local Replication	Remote Replication	Device Management
Create/delete user accounts	No	No	No

Table 6. Permissions for Local Replication, Remote Replication, and Device Management roles (continued)

Permissions	Local Replication	Remote Replication	Device Management
Reset user password	No	No	No
Create roles	No	No	No
Change own password	Yes	Yes	Yes
Manage storage systems	No	No	No
Discover storage systems	No	No	No
Add/show license keys	No	No	No
Set alerts and Optimizer monitoring options	No	No	No
Release storage system locks	No	No	No
Set Access Controls	No	No	No
Set replication and reservation preferences	No	No	No
View the storage system audit log	No	No	No
Access performance data	Yes	Yes	Yes
Start data traces	Yes	Yes	Yes
Set performance thresholds/ alerts	No	No	No
Create and manage performance dashboards	Yes	Yes	Yes
Perform control and configuration operations on devices	No	No	Yes
Create, expand, or delete devices	No	No	No
Perform local replication operations (SnapVX, legacy Snapshot, Clone, BCV)	Yes	No	No
Create Secure SnapVX snapshots	No	No	No
Create, operate upon, or delete SRDF device pairs	No	Yes	No
Create, modify, or delete SRDF groups	No	No	No

Create local users

Users gain access to a storage system or component either directly through a role assignment or indirectly through membership in a user group that has a role assignment.

Prerequisites

To perform this operation, you must be the Initial Setup User (set during installation), or SecurityAdmin on at least one storage system.

About this task

See Roles and associated permissions for an overview of the Role-Based Access Control (RBAC) functionality.

This procedure explains how to create local users. Local users have accounts that are stored locally in the user database in the Unisphere server host (Log in to Unisphere with your Username and Password.).

Steps

1. Select  to open the **Settings** panel.
2. Select **Users and Groups** > **Local Users.**
3. Click **Create** to open the **Create Local User** dialog box.
4. Type a **User Name**.

   User names are case-sensitive and allow alphanumeric characters of either case, an underscore, a dash, a period, or a $:

   - a-z
   - A-Z
   - 0-9
   - _
   - .
   - -
   - $
5. Optional: Type a **Description**.

   The text within the description supports the following characters:

   - a-z
   - A-Z
   - 0-9
   - _
   - -
   - .
   - !
   - @
   - Number
   - $
   - %
   - &
   - *
   - (
   -)
   - .
   - ^
6. Type and confirm a user **Password**. Click the eye icon when typing in the password and you can see what you are entering.

   Passwords cannot exceed 16 characters. There are no restrictions on special characters when using passwords.
7. On the **Roles** tab, select the object and up to four roles.
8. If you choose a **Local Replication**, **Remote Replication** or **Device Management** role, click **Select Storage Group(s)** and in the edit dialog that opens choose between:
   a. **Wildcard**—A wildcard syntax used with the storage group component name to allow a single rule to apply to multiple storage groups.

      A simple wildcard syntax can be used with the component name to allow a single rule to apply to multiple SGs as follows:

      abc - Exactly these characters

      ? Any one character

      * Any zero or more characters

      + Zero or more additional occurrences of the previous match

      [a-z0-9] Any of these characters

[!a-z] Anything but one of these characters

All SG name comparisons are case-insensitive. The following table contains examples to show how they are interpreted:

Table 7. Wildcard syntax examples

This pattern	Matches these Storage Groups	Does not match these Storage Groups
tg_*	tg_DB_SG1 or tg_newSG or TG_sg_db	tgNewSG
prod_sg?	prod_sg1 or prod_sga or Prod_sg2	prod_sg12 or prod_sgab
prod_sg[0-9]+	prod_sg1 or prod_sg12	prod_sga or prod_sgab

The only allowed characters are: a-zA-Z0-9_- along with the above *+?[]! wildcard characters.

The only roles that can be assigned against storage groups are: Local Replication, Remote Replication, and Device Management.

Storage groups do not have to exist at the time that a matching Role-Based Authentication Controls (RBAC) rule for them is defined.

These storage group-level RBAC rules are only applicable to parent and stand-alone SGs and not child SGs. Child SGs are protected by the RBAC rules, if any, on their parent SG.

b. **Storage Group**
c. Once your input or selection is complete, click **Save**.

9. Click **OK**.

Roles for performing local and remote replication actions

Users gain access to a storage system or component either directly through a role assignment and/or indirectly through membership in a user group that has a role assignment.

The following table details the roles that are required to perform local and remote replication actions.

NOTE: Multiple Yesses imply that both are needed, for example, **Link Snapshot** requires the user to have Local Replication rights on the snapshot device and Device Manager rights on the link device.

NOTE: Unisphere for PowerMax does not support RBAC device group management.

Table 8. Permissions for Local Replication, Remote Replication, and Device Management roles

Permissions	Local Replication	Remote Replication	Device Manager
Protection Wizard - Create SnapVx Snapshot	Yes (a)	No	No
Create Snapshot	Yes (a)	No	No
Edit Snapshot	Yes	No	No
Link Snapshot	Yes (b) (c)	No	Yes (d)
Relink Snapshot	Yes (b) (c)	No	Yes (d)
Restore Snapshot	Yes (b)	No	Yes (b)
Set Time To Live	Yes	No	No
Set Mode	Yes (b)	No	Yes (d)
Terminate Snapshot	Yes	No	No
Unlink Snapshot	Yes (b)	No	Yes (d)
SRDF Delete	No	Yes	No
SRDF Establish	No	Yes	No

Table 8. Permissions for Local Replication, Remote Replication, and Device Management roles (continued)

Permissions	Local Replication	Remote Replication	Device Manager
SRDF Failback	No	Yes	No
SRDF Failover	No	Yes	No
SRDF Invalidate	No	Yes	No
SRDF Move	No	Yes	No
SRDF Not Ready	No	Yes	No
SRDF R1 Update	No	Yes	No
SRDF Ready	No	Yes	No
SRDF Refresh	No	Yes	No
SRDF Restore	No	Yes	No
SRDF Resume	No	Yes	No
SRDF RW Disable R2	No	Yes	No
SRDF RW Enable	No	Yes	No
SRDF Set Bias	No	Yes	No
SRDF Set Consistency	No	Yes	No
SRDF Set Mode	No	Yes	No
SRDF Set SRDF/A	No	Yes	No
SRDF Split	No	Yes	No
SRDF Suspend	No	Yes	No
SRDF Swap	No	Yes	No
SRDF Write Disable	No	Yes	No

(a) - Set Secure is blocked for users who only have Local_REP rights.

(b) - The user must have the specified rights on the source volumes.

(c) - The user may only choose existing storage groups to link to. Creating a storage group requires Storage Admin rights.

(d) - The user must have the specified rights on the link volumes.

RBAC roles for SRDF local and remote replication actions

A user must be assigned the necessary roles to perform SRDF local and remote replication actions.

The following table details the roles that can perform SRDF local and remote replication actions:

NOTE: Unisphere for PowerMax does not support RBAC device group management.

Table 9. Permissions for Local Replication, Remote Replication, and Device Management roles

Permissions	Local Replication	Remote Replication	Device Manager
SRDF Delete	No	Yes	No
SRDF Establish	No	Yes	No
SRDF Failback	No	Yes	No
SRDF Failover	No	Yes	No
SRDF Invalidate	No	Yes	No

Table 9. Permissions for Local Replication, Remote Replication, and Device Management roles (continued)

Permissions	Local Replication	Remote Replication	Device Manager
SRDF Move	No	Yes	No
SRDF Not Ready	No	Yes	No
SRDF R1 Update	No	Yes	No
SRDF Ready	No	Yes	No
SRDF Refresh	No	Yes	No
SRDF Restore	No	Yes	No
SRDF Resume	No	Yes	No
SRDF RW Disable R2	No	Yes	No
SRDF RW Enable	No	Yes	No
SRDF Set Bias	No	Yes	No
SRDF Set Consistency	No	Yes	No
SRDF Set Mode	No	Yes	No
SRDF Set SRDF/A	No	Yes	No
SRDF Split	No	Yes	No
SRDF Suspend	No	Yes	No
SRDF Swap	No	Yes	No
SRDF Write Disable	No	Yes	No

RBAC roles for TimeFinder SnapVX local and remote replication actions

A user must be assigned the necessary roles to perform TimeFinder SnapVX local and remote replication actions.

The following table details the roles that are required to perform TimeFinder SnapVX local and remote replication actions:

NOTE: Multiple Yesses imply that both are needed, for example, **Link Snapshot** requires the user to have Local Replication rights on the snapshot device and Device Manager rights on the link device.

NOTE: Unisphere for PowerMax does not support RBAC device group management.

Table 10. Permissions for Local Replication, Remote Replication, and Device Management roles

Permissions	Local Replication	Remote Replication	Device Manager
Protection Wizard - Create SnapVx Snapshot	Yes [a]	No	No
Create Snapshot	Yes [a]	No	No
Edit Snapshot	Yes	No	No
Link Snapshot	Yes [b] [c]	No	Yes [d]
Relink Snapshot	Yes [b] [c]	No	Yes [d]
Restore Snapshot	Yes [b]	No	Yes [b]
Set Time To Live	Yes	No	No
Set Mode	Yes [b]	No	Yes [d]

Table 10. Permissions for Local Replication, Remote Replication, and Device Management roles (continued)

Permissions	Local Replication	Remote Replication	Device Manager
Terminate Snapshot	Yes	No	No
Unlink Snapshot	Yes (b)	No	Yes (d)

(a) - Set Secure is blocked for users who only have Local_REP rights.

(b) - The user must have the specified rights on the source volumes.

(c) - The user may only choose existing storage groups to link to. Creating a storage group requires Storage Admin rights.

(d) - The user must have the specified rights on the link volumes.

Understanding workload monitoring, planning, and provisioning

Unisphere supports applications that support the monitoring, planning, and provisioning of workloads.

If you have an array that is monitored for performance by Unisphere for PowerMax for at least two weeks you have enough data to do a characterization of the workload for an existing storage group or multiple storage groups.

The workload planner (WLP) application performs performance impact checks (the terms performance impacts checks, admissibility checks, and suitability checks are used interchangeably) when provisioning using the Unisphere UI wizards; it performs the checks to ensure that there is sufficient performance headroom before creating application workloads on storage arrays. WLP keeps its own database which is a subset of Key Performance Indicator (KPI) metrics that Unisphere for PowerMax gathers, and uses this database to help make decisions, so to use it the arrays in question must be registered for performance data collection.

WLP models the internal load balancing of the array when adding additional load to an existing array. WLP can help in placing workloads on the most suitable target array for new provisioning and/or migrated workloads. WLP can help in port group creation or selection that is based on existing zoning and utilization, while balancing across directors.

A Note on Data Processing and Organization

Many applications (and as a result, storage systems) experience fluctuations in workloads over time. Often, these fluctuations repeat in a predictable way. In order to keep the Workload Planner data footprint as small as possible, but still adequately represent the history and fluctuations in workloads, multiple weeks of historical storage system performance data are weighted and averaged into a week's worth of four-hour time windows which create a "representative workload" for all system components and applications that are applied to a storage system.

Best practice limits are applied when accounting for the maximum performance capabilities of a system component. All performance impact scores and utilizations that Workload Planner calculates and reports are against these best practice limits, not the actual maximum performance capacity of the pertinent component or components. Recommendations and performance impact scores are intentionally conservative to ensure that maximum system performance is maintained.

Also, all utilization and performance impact scores are derived from the four-hour windows. When a single value is returned, the value return is the value of the most performance-limiting four-hour window value for the most limiting component on the storage system. For example, during new provisioning, the impact of adding a workload to a front-end port could take the utilization of that port from ~30% used to ~35% for most time windows, but 32% to 37% on Tuesdays between 8AM and 12PM UTC. In this case, the new port utilization would be reported as 37% because that is the most limiting four-hour window value.

The four-hour window implementation allows Workload Planner to pack workloads more efficiently and realistically onto storage systems regarding inverse performance fluctuations. For example, a workload that is heavy on weekends, but light during weekdays performs better with a second workload that is light on weekends and heavy on weekdays than a workload that is distributed similarly to itself.

One drawback to the four-hour window implementation is that it is sensitive to peaks in data. Data Exclusion Windows are used to resolve unexpected performance impact, system headroom, and/or service level compliance results.

The following workload monitoring, planning, and provisioning tasks can be performed from Unisphere for PowerMax:

- Monitor Storage Group compliance against its service level and workload type:
  - Create service level compliance alert policies.
  - View storage group compliance
  - View compliance report
  - Create a compliance report.

- Exclude data.
- Perform provisioning.
- Perform multi-array provisioning.
- Perform multi-array provisioning using templates.
- Perform migration.
- Perform multi-array migration.

Managing planning jobs using a Planning dashboard

Unisphere for PowerMax provides a planning dashboard that allows users to see historical and predicted headroom information and this information can be used to create and manage planning jobs.

Prerequisites

- See Roles and associated permissions to determine the user roles that can perform this task.

About this task

Unisphere provides a new planning dashboard, displaying historical and predicted headroom data for each array, helping you make more informed decisions. The new progressive provisioning wizard allows you to quickly determine the best location for a new workload, considering the total system performance of all available candidate storage systems. You can define workloads with minimal input for a rapid suitability check or follow a guided process with automatic port selection for a more detailed configuration. Once you have defined a planning job, you can see the impact of the new workload on the headroom of the array over time, enabling you to plan for optimal performance.

Headroom is the amount (in TB) of a typical workload that can be added to a system without exceeding physical capacity or component performance capacity. Instead of displaying a single current headroom value for each storage system, historical and projected future headroom values are displayed in charts on the planning dashboard. As planning jobs are introduced, you can see how these jobs reduce the remaining projected headroom on each storage system managed by an instance of Unisphere for PowerMax.

The planning dashboard supports new provisioning. The new workflow provides simplified and guided configuration recommendations and dynamically updated performance impact results.

A new metric called storage group impact is calculated and displayed. This value represents the relative impact that each storage group has on its current storage system. A higher value represents a higher impact. This value can be used to identify potential migration candidates.

Steps

1. Click **Planning** from the side panel on the home dashboard.
2. On the Planning dashboard, select a time period and view the following information for each storage system:
   - A dashboard displaying headroom charts that display over time the historical remaining headroom and the predicted remaining headroom. The headroom charts for the different systems are displayed in a single section (if no planning job is present) or in two different sections based on which systems are affected by the planning job and which remain unaffected (if a planning job has been created). Click **typical workload** to see the typical workload details,
   - For a selected headroom chart, a new metric called storage group impact is calculated and displayed. This value represents the relative impact that each storage group has on its current storage system. A higher value represents a higher impact. This value can be used to identify potential migration candidates.
3. Select a storage system by clicking the headroom data card of the storage system and viewing the **Storage Group Impacts** values in the **Details** panel.
   The **Planning** dashboard can be in one of the following states:
   - A planning dashboard with no existing planning job. Click **Provision**.
   - A planning dashboard with a saved Planning Job in a **Ready to Run** state. Go to step 12.
   - A planning dashboard with a saved Planning Job in an **Invalid** state. Click **Replan**.

   NOTE: If the saved planning job is not run or not added to the job list, it is possible (based on other changes to the environment) that the selected systems may no longer be the recommended systems and the saved job may transition to an **Invalid** state.
4. Specify the required capacity.
5. Specify the workload details.

6. Specify the workload performance by specifying the IOPS or by using the workload from a selected provisioning template (that has been previously created) and click **OK**.
7. Specify the hosts and ports.

   A list of recommended (suitable) systems is displayed. Unsuitable systems are also displayed.
8. For a recommended system, click the arrow button next to **Select** to see a diagram summarizing performance impact and configuration information.
9. Click **Select** to select one of the recommended systems and click **Next**.
10. A **Customize** page is displayed that has a diagram summarizing performance impact and configuration information . The associated configuration and status information is displayed in the following tabs:
    - **Next Steps**—The next steps are listed, for example, are you ready to save the planning job.
    - **Hosts**—The related hosts and their associated configuration details are displayed.
    - **Masking Views**—The related masking views and their associated configuration details are displayed.
    - **Initiators**—The related initiators and their associated configuration details are displayed.
    - **Port Groups**—The related port groups and their associated configuration details are displayed. Click a port group to view additional configuration details. Click **Modify** to edit a port group.
    - **Storage Groups**—The related storage groups and their associated configuration details are displayed.
11. Review the **Customize** page. Optionally click **View Provisioning Settings** and **Details** to view detailed information. Click **Back** if you want to make any changes or click **Save Planning Job** to go to the next step.
12. The planning dashboard is displayed and shows the details of the saved planning job. You can delete the planning job by hovering over the row and clicking  . You can modify the planning job details by selecting the job. If you want to proceed without changing anything, do one of the following:
    - Click **Run Now** to perform the operation now.
    - Expand **Run Now** and click **Add to Job List** to add this task to the job list, from which you can schedule or run the task at your convenience.

Storage Management

Storage Management covers Unisphere for PowerMax support for the management of storage within one or more storage systems.

Storage Management covers the following areas:

- Storage Group management - Storage groups are a collection of devices that are stored on the array, and an application, a server, or a collection of servers use them.
- Service Level management - A service level is the response time target for a storage group. The service level sets the storage system with the required response time target for a storage group. It automatically monitors and adapts to the workload needed to maintain the response time target. The service level includes an optional workload type so it can be optimized to meet performance levels.
- Template management - Using the configuration and performance characteristics of an existing storage group as a starting point, you can create templates that will pre-populate fields in the provisioning wizard and create a more realistic performance reservation in your future provisioning requests.
- Storage Resource Pool management - SRP management provides automated management of storage system disk resources to achieve expected service levels. Disk groups can be configured to form a Storage Resource Pool (SRP) by creating thin pools according to each individual disk technology, capacity, and RAID type.
- Volume management - A storage volume is an identifiable unit of data storage. Storage groups are sets of volumes.
- External Storage management - Attaching external storage to storage systems directs workload movement to these external storage systems while having access to the storage system features such as local replication, remote replication, storage tiering, data management, and data migration. Also, it simplifies multi-vendor or Dell storage system management.
- vVol management - VMware vVols enable data replication, snapshots, and encryption to be controlled at the VMDK level instead of the LUN level, where these data services are performed on a per VM (application level) basis from the storage array.
- Disk Groups management - A disk group is a collection of hard drives within the storage system that share the same performance characteristics.
- File - Configure File Systems.

Understanding storage groups

Storage groups are a collection of devices that are stored on the array. An application, a server, or a collection of servers use them.

- The maximum number of storage groups that are allowed on a storage system is 16,384.
- The maximum number of child storage groups that are allowed in a cascaded configuration is 128.
- A storage group can contain up to 4,096 volumes.
- A volume can belong to multiple storage groups when only one of the groups has an SRP.
- You cannot create a storage group containing CKD volumes and FBA volumes.

Understanding data reduction

Data reduction allows users to reduce user data on storage groups and storage resources.

Data reduction is enabled by default and can be turned on and off at storage group and storage resource level.

If a storage group is cascaded, enabling data reduction at this level enables data reduction for each of the child storage groups. The user can disable data reduction on one or more of the child storage groups.

To turn the feature off on a particular storage group or storage resource, clear the **Enable Data Reduction** check box in the **Create Storage Group**, **Modify Storage Group,** or **Add Storage Resource To Storage Container** dialogs or when using the **Provision Storage** or **Create Storage Container** wizards.

The following are the prerequisites for using data reduction:

- Data reduction is only allowed on All Flash systems running the HYPERMAX OS 5977 Q3 2016 Service Release, or PowerMaxOS 5978, or greater.
- Data reduction is allowed on FBA devices.
- Data reduction is allowed on CKD devices for storage systems running PowerMaxOS 10 (6079).
- The user must have at least StorageAdmin rights.
- The storage group must have an associated SRP.
- The associated SRP cannot be comprised, either fully or partially, of external storage.

Reporting

Users can see the current data reduction ratio on the device, the storage group, and the SRP. Efficiency ratios are reported in units of 1/10:1.

NOTE: External storage is not in efficiency reports. Mixed SRPs with internal and external storage, only the internal storage is used in the efficiency ratio calculations.

Understanding service levels

A service level is the response time target for the storage group. The service level enables you set the storage system with the required response time target for the storage group.

It automatically monitors and adapts to the workload to maintain (or meet) the response time target. The service level includes an optional workload type. The optional workload type can be used to further tune expectations for the workload storage group to provide enough flash to meet your performance objective.

Performance Impact

The performance impact operation determines whether the storage system can handle the updated service level.

The performance impact is only available when:

- The storage system is registered with the performance data processing option for statistics.
- The workloads have been processed.
- All the storage groups that are involved have a service level and SRP set.

- The target SRP does not contain only external disk groups.
- The storage system is local.
- The storage group is not in a masking view (only for the local provisioning wizard).
- An issue with one of the selected ports arises when provisioning storage and a valid Front-End Suitability score cannot be derived. Examples of issues are: A virtual port is selected, an offline port is selected, and a selected port has no negotiated speed. When an issue arises, 200.0% (not a real suitability score) is displayed. Excluding data has no impact on the 200% displayed value.

This message indicates whether the storage system can handle the updated service level. Results are indicated with either of the following:

-  Indicates suitable.
-  Indicates not suitable.

In both cases, results are displayed in a bar chart by component (Front End, Back End, Cache) along with a score 0 to 100. The score can be viewed by hovering the cursor over the bar. It indicates the expected availability of the components on the target storage system after the change.

The current score for the component is shown in gray, with the additional load for the component shown in green or red indicating suitability. The additional score is red if the current and additional loads total more than 100.

Understanding storage templates

Storage templates are a reusable set of storage requirements that simplify storage management for virtual data centers by eliminating many of the repetitive tasks that are required to create and make storage available to hosts or applications.

With this feature, Administrators and Storage Administrators can create templates for their common provisioning tasks and then invoke them later when performing such things as:

- Creating or provisioning storage groups.

The templates that are created on a particular Unisphere server can be used across all the arrays on that particular server.

A provisioning template contains configuration information and a performance reservation.

The performance reservation that is saved with a template is generated from a 2-week snapshot of the performance data of the source storage group. The total IOPS and MBPS, I/O mixture, and skew profile from this snapshot are used for array impact tests when the template is used to provision a new storage group.

Understanding Storage Resource Pools

A Storage Resource Pool is a collection of data pools that are used for capacity and performance management.

By default, a single default Storage Resource Pool is pre-configured at the factory. More Storage Resource Pools can be created with a service engagement.

Understanding volumes

A storage volume is an identifiable unit of data storage. Storage groups are sets of volumes.

The Volumes view on the Unisphere for PowerMax user interface provides you with a single place from which to view and manage all the volume types on the system.

Understanding disk groups

A disk group is a collection of hard drives within the storage system that share the same performance characteristics.

Disk groups can be viewed and managed from the Unisphere for PowerMax user interface.

Host Management

Storage hosts are systems that use storage system LUN resources. Unisphere manages the hosts.

Host Management covers the following areas:

- Management of host and host groups
- Management of masking views - A masking view is a container of a storage group, a port group, and an initiator group , and makes the storage group visible to the host. Devices are masked and mapped automatically. The groups must contain some devices entries.
- Management of port groups - Port groups contain director and port identification and belong to a masking view. Ports can be added to and removed from the port group. Port groups that are no longer associated with a masking view can be deleted.
- Management of initiators and initiator groups - An initiator group is a container of one or more host initiators (Fibre or iSCSI). Each initiator group can contain up to 64 initiator addresses or 64 child IG names. Initiator groups cannot contain a mixture of host initiators and child IG names.
- Management of PowerPath hosts
- Management of mainframe configured splits, CU images, and CKD volumes

Understanding hosts

Storage hosts are systems that use storage system LUN resources. A logical unit number (LUN) is an identifier that is used for labeling and designating subsystems of physical or virtual storage.

The maximum number of initiators that is allowed is 64.

Understanding masking views

A masking view is a container of a storage group, a port group, and an initiator group , and makes the storage group visible to the host.

Masking views are manageable from the Unisphere for PowerMax user interface. Devices are masked and mapped automatically. The groups must contain some devices entries.

Understanding port groups

Port groups contain director and port identification and belong to a masking view. Ports can be added to and removed from the port group. Port groups that are no longer associated with a masking view can be deleted.

Note the following recommendations:

Port groups should contain four or more ports.

Each port in a port group should be on a different director.

A port can belong to more than one port group. However, you cannot mix different types of ports (physical FC ports, virtual ports, and iSCSI virtual ports) within a single port group.

Understanding initiators

An initiator group is a container of one or more host initiators (Fibre or iSCSI or NVMe/TCP).

Each initiator group can contain up to 64 initiator addresses or 64 child IG names. Initiator groups cannot contain a mixture of host initiators and child IG names.

Understanding PowerPath hosts

PowerPath is host-based software that provides automated data path management and load-balancing capabilities for heterogeneous server, network, and storage deployed in physical and virtual environments.

The following are the minimum requirements to perform this task:

- A storage system running PowerMax 5978 or higher
- PowerPath 6.3

Understanding mainframe management

Service level provisioning for mainframe simplifies storage system management by automating many of the tasks that are associated with provisioning storage.

The mainframe dashboard provides you with a single place to monitor and manage configured splits, CU images, and CKD volumes.

In 10.0, SE blocks the following operations that are triggered from a U4P server that is installed on a mainframe host - z/OS and z/Linux hosts:

- Map/unmap to/from CU
- Create/delete CU on all z/OS hosts
- Create/delete CU on all z/Linux hosts

One of the properties of the CKD volume is the physical name (also known as volume serial number (VOLSER)). VOLSER information is displayed if running a SMAS client/server to zOS as described in Appendix C - Configuring SMAS to work in z/OS of the *Unisphere for PowerMax Installation Guide*.

The mainframe dashboard is organized into the following panels:

- **CKD Compliance** - Displays how well CKD storage groups are complying with their respective service level policies, if applicable.
- **CKD Storage Groups** - Displays the mainframe storage groups on the storage system. Double-click a storage group to see more details and information about its compliance and volumes.
- **Actions** - Allow the user to provision storage and create CKD volumes.
- **Summary** - Displays the mainframe summary information in terms of splits, CU images, and CKD volumes.

With the release of HYPERMAX OS 5977 Q1 2016, Unisphere introduces support for service level provisioning for mainframe. Service level provisioning simplifies storage system management by automating many of the tasks that are associated with provisioning storage.

Service level provisioning eliminates the need for storage administrators to manually assign physical resources to their applications. Instead, storage administrators specify the service level and capacity that is required for the application and the system provisions the storage group appropriately.

You can provision CKD storage to a mainframe host using the Provision Storage wizard.

The storage system must be running HYPERMAX OS 5977 Q1 2016, or higher, and have at least one FICON director configured.

You can map CKD devices to front-end EA/EF directors. Addressing on EA and EF directors is divided into Logical Control Unit images, also known as CU images. Each CU image has its own unique SSID and contains a maximum of 256 devices (numbered 0x00 through 0xFF). When mapped to an EA or EF port, a group of devices becomes part of a CU image.

With the release of HYPERMAX OS 5977 Q2 2017, Unisphere introduces support for All Flash Mixed FBA/CKD storage systems.

NOTE: This feature is only available for All Flash 450F/850F/950F storage systems that are:

- Purchased as a mixed All Flash system
- Installed at HYPERMAX OS 5977 Q2 2017 or later
- Configured with two Storage Resource Pools - one FBA Storage Resource Pool and one CKD Storage Resource Pool

You can provision FBA/CKD storage to a mainframe host using the **Provision Storage** wizard.

NOTE:

1. A CKD SG can only provision from a CKD SRP.
2. A FBA SG can only provision from a FBA SRP.

3. FBA volumes cannot reside in a CKD SRP.
4. CKD volumes cannot reside in a FBA SRP.
5. Compression is only for FBA volumes.

You can map FBA devices to front-end EA/EF directors. Addressing on EA and EF directors is divided into Logical Control Unit images (CU images). Each CU image has its own unique SSID and contains a maximum of 256 devices (numbered 0x000 through 0xFF). When mapped to an EA or EF port, a group of devices becomes part of a CU image.

Manage storage for mainframe

The mainframe dashboard provides you with a single place to monitor and manage configured splits, CU images, volumes, and storage groups.

About this task

To access the mainframe dashboard:

Steps

1. Select the storage system.
2. Select **Hosts > Mainframe** to open the mainframe dashboard.

   The mainframe dashboard is organized into the following areas:

   - **Directors** panel
   - **Navigable Links** panel
   - **SRDF CKD Relationships** panel
   - **CKD SRP Effective Capacity Usage** panel
   - **Action Links** panel
   - **System Snapshot Usage** tab
   - **System Provisioned Capacity** tab
   - **SRDF CKD Group Status** panel

   Directors

   The **Directors** panel displays icons representing the status of directors under **Performance** and **Hardware** headings.

   To open the **Performance** dashboard for that director type, click the director type in the **Performance** section.

   To open the hardware list of directors for that director type, click the director type in the **Hardware** section.

   Navigable Links

Performance	The following performance data is displayed: • Host IOs • Latency • Throughput To open the **CKD** tab of the **System Performance** dashboard, click the **Performance** heading.
CKD Protection	The **CKD Protection** section has graphics displaying the number and status of CKD storage groups. To go to the **CKD** tab of the Protection dashboard, click the **CKD Protection** heading.
Mainframe Objects	The **Mainframe Objects** section lists the total number of each of the following mainframe objects: • Splits • CU Images • Volumes • Storage Groups

   To open the list view for any of these objects, click the number of objects.

   SRDF CKD Relationships

   The **SRDF CKD Relationships** panel displays a topology graphic showing the CKD SRDF relationship information. The color of the link indicates the status of the SRDF pair:

- Green: Synchronized, Sync in Progress, Consistent
- Yellow: Suspended, Split, Failed Over, R1 Updated, R1 Update in Progress, Invalid
- Red: Partitioned, Transmit Idle

A tooltip displays the number of SRDF groups with each status, and each mode (Synchronous, Adaptive Copy, and Asynchronous).

CKD SRP Effective Capacity Usage

The **CKD SRP Effective Capacity Usage** panel displays a graphic showing the following information:

- Total effective capacity
- Percentage used
- Capacity used by snapshot
- Capacity used by user
- Saved capacity
- Free capacity

To go to the **Capacity** dashboard, click the **CKD SRP Effective Capacity Usage** heading.

Action Links

The Action Links panel displays the following links:

Provision Storage	Opens the **Mainframe Provision** wizard, which guides you through the process of provisioning storage for a mainframe.
Create CKD Volumes	Opens the **Create Volume** dialog, from where you can create a CKD volume.

System Snapshot Usage

The **System Snapshot Usage** tab displays a graphic showing the amount of snapshot capacity that is used and how close the system is to breaching its threshold.

System Provisioned Capacity

The **System Provisioned Capacity** tab displays a graphic showing the amount of provisioned capacity that is used and how close the system is to breaching its threshold.

SRDF CKD Group Status

The **SRDF CKD Group Status** displays the following information about the status of CKD SRDF groups:

- Name—Name of the CKD group
- Status—Status indicated by an icon:
  - Green: Synchronized, Synchronization in Progress, Consistent
  - Yellow: Suspended, Split, Failed Over, R1 Updated, R1 Update in Progress, Invalid
  - Red: Partitioned, Transmit Idle
- Modes—Mode of the CKD group, for example, Synchronous, Adaptive Copy, Asynchronous
- Emulation—Emulation used by the CKD group
- Online—Indicates if the CKD group is online.

Data protection management

Data Protection management ensures that data is protected and remains available.

Data Protection Management covers the following areas:

- Management of snapshot policies
- Management of MetroDR
- Management of SRDF groups - SRDF groups provide a collective data transfer path linking volumes of two separate storage systems. These communication and transfer paths are used to synchronize data between the R1 and R2 volume pairs that are associated with the SRDF group. At least one physical connection must exist between the two storage systems within the fabric topology. See the *Dell SRDF Introduction Guide* for an overview of SRDF.
- Migrations management - enables the migration of storage group (application) data from migration capable source arrays to migration capable target arrays.

- Management of Virtual Witness - The Witness feature supports a third party that the two storage systems consult when they lose connectivity with each other, that is, their SRDF links go out of service. When SRDF links go out of service, the Witness helps to determine, for each SRDF/Metro Session, which of the storage systems should remain active (volumes continue to be read and write to hosts) and which goes inactive (volumes not accessible).
- Management of Open Replicator - Open Replicator is a software application that is used to migrate data from third-party arrays to PowerMax arrays.
- Management of device groups - A device group is a user-defined group consisting of devices that belong to a locally attached array. Control operations can be performed on the group as a whole, or on the individual device pairs in the group. By default, a device can belong to more than one device group.
- File Protection - Protect File Systems (optionally protect file system data using snapshots or replication).
- File Migration - Migration support is provided for file objects, system configuration, and user data from eNAS to PowerMax File.

Manage SRDF storage groups

As part of the AIOPS Predictive and Prescriptive Remediation initiative, Unisphere for PowerMax provides an overall view of the replication status of a storage group. This view provides a visual and interactive means of monitoring the replication from the viewpoint of an application. The focus is on providing users with sufficient information about each SRDF storage group (the storage group and its configured replication solution) so that they can make informed decisions more quickly and without the need to collate information from throughout Unisphere for PowerMax.

Prerequisites

See Roles and associated permissions to determine the user roles that can perform this task.

About this task

The SRDF Storage Group Details View provides you with a high-level overview of the SRDF status of a storage group, and is specially geared to improve your experience with HA/DR replication solutions. The high-level diagram helps to visualize how the components that make up the PowerMax HA/DR solutions interact with one another and the context they operate in. From this view, you can quickly understand any current issues affecting a SRDF storage group that could impact its ability to replicate data. This view also mirrors several list views from throughout Unisphere for PowerMax that augment the information that is shown and provide useful actions.

The SRDF Storage Group Details view provides specific warnings and help for a number of issues that may impact the performance of the replication configuration of the user.

The Critical state is determined if any of the following is true:

- All initiators that are used in a storage group associated masking view are not logged in.
- All initiators that are used in a storage group associated masking view are not on fabric.
- All front end ports in a port group are offline.
- All OR ports enabled for SRDF (V4) or all RF/RE ports enabled for SRDF (V3) are offline.
- The remote storage group in an SRDF/A configuration is more than 99 cycles behind the local storage group.
- The SRDF state of any SRDF pair in the SRDF Storage Group is in the **Partitioned** state.
- The SRDF state of any SRDF pair in the SRDF Storage Group is in the **Transmit Idle** state.
- The SRDF state of any SRDF pair in the SRDF Storage Group is in the **Failed** state.

The Warning state is determined if any of the following is true:

- Some of the initiators that are used in a storage group associated masking view are not logged in.
- Some of the initiators that are used in a storage group associated masking view are not on fabric.
- Some of the front end ports in a port group are offline.
- Some OR ports enabled for SRDF (V4) or some RF/RE ports enabled for SRDF (V3) are offline.
- The remote storage group in an SRDF/A configuration is more than three cycles behind and fewer than 100 cycles behind the local storage group.
- The storage group is protected by Metro, but does not have a vWitness configured.
- The SRDF state of any SRDF pair in the SRDF Storage Group is in the **Suspended** state.
- The SRDF state of any SRDF pair in the SRDF Storage Group is in the **Split** state.
- The SRDF state of any SRDF pair in the SRDF Storage Group is in the **Failed Over** state.
- The SRDF state of any SRDF pair in the SRDF Storage Group is in the **R1 Update in Progress** state.
- The SRDF state of any SRDF pair in the SRDF Storage Group is in the **R1 updated** state.

- The SRDF state of any SRDF pair in the SRDF Storage Group is in the **Unknown** state.
- The SRDF state of any SRDF pair in the SRDF Storage Group is in the **Invalid** state.
- The DSE Pool is active for the SRDF group.

Steps

1. Select the storage system.
2. Select **Data Protection > Manage SRDF SGs**.

   A list view is displayed that allows you to see the list of SRDF Storage Groups that have been configured for remote replication. The view has three tabs - Critical, Warning, and Normal. The Critical tab is displayed by default. In each tab, a list of storage groups is displayed. The following properties display for each storage group:

   - **Storage Groups**—Name of the storage group
   - **SRDF Configuration**—SRDF Configuration type (**Single**, **Metro**, **Cascaded**, **Concurrent**, **MetroDR**, and **Metro with DR**)
   - **Issues**—Issues associated with the storage group
3. Select a tab.
4. Select the checkbox next to a storage group to view a summary of the overall replication status of the storage group.
5. Click the storage group name. For a storage group with a single replication path (**Single**), the SRDF Status Details View containing the associated configuration and status information is displayed. For a storage group with multiple replication paths (**Cascaded**, **Concurrent**, **Metro**, **MetroDR** and **Metro with DR**), a diagram showing all the replication paths is displayed. Select a replication path to view the SRDF Status Details View containing the associated configuration and status information. The associated configuration and status information is displayed in the following tabs:
   - **Overview**—An overview of the SRDF status is displayed.
   - **Hosts**—The related hosts and their associated configuration details are displayed. Click a host to view additional configuration details. Click **Manage Hosts** to manage the hosts.
   - **Masking Views**—The related masking views and their associated configuration details are displayed. Click a masking view to view additional configuration details. Click **Manage Masking Views** to manage the masking views.
   - **Initiators**—The related initiators and their associated configuration details are displayed. Click an initiator to view additional configuration details. Click **Manage Initiators** to manage the initiators.
   - **Port Groups**—The related port groups and their associated configuration details are displayed. Click a port group to view additional configuration details. Click **Manage Port Groups** to manage the port groups.
   - **Hardware**—The related hardware directors and their associated status details are displayed. Click a director to view additional configuration details. Click **Online** to set a selected port to be online. Click **Offline** to set a selected port to be offline. Click **Manage OR Directors** to manage the OR directors. Click **Manage EF Directors** to manage the EF directors. Click **Manage FE Directors** to manage the FE directors. Click **Manage RDF Directors** to manage the SRDF directors.
   - **SRDF Groups**—The related SRDF groups and their associated configuration details are displayed. Click a SRDF group to view additional configuration details. Click **Manage SRDF Groups** to manage the SRDF groups.
   - **Data Protection**—The storage group SRDF relationships information is displayed. Click a storage group to view additional configuration details. Click **Manage Data protection** to manage the storage group SRDF relationships.

Understanding Snapshot policy

The Snapshot policy feature provides snapshot orchestration at scale (1,024 snaps per storage group). The feature simplifies snapshot management for snapshots.

The flowchart below depicts the basic workflow of using snapshot policies:



Snapshots can be used to recover from data corruption, accidental deletion, or other damage, offering continuous data protection. Many snapshots can be difficult to manage. The Snapshot policy feature provides an end to end solution to create, schedule, and manage snapshots.

The snapshot policy (Recovery Point Objective (RPO)) specifies how often the snapshot should be taken and how many of the snapshots should be retained. The snapshot may also be specified to be secure (these snapshots cannot be terminated by users

before their time to live (TTL), derived from the snapshot policy's interval and maximum count, has expired). Up to four policies can be associated with a storage group, and a snapshot policy may be associated with many storage groups. Unisphere provides views and dialogs to view and manage the snapshot policies. Unisphere also calculates and reports on the compliance of each storage group to its snapshot policies.

The following rules apply to snapshot policies:

- The maximum number of snapshot policies that can be created on a storage system is 20. Multiple storage groups can be associated with a snapshot policy.
- A maximum of four snapshot policies can be associated with an individual storage group.
- A storage group or device can have a maximum of 256 manual snapshots.
- A storage group or device can have a maximum of 4000 snapshots.
- When there are 4000 snapshots in existence and another snapshot is taken, the oldest unused snapshot that is associated with the snapshot policy is removed.
- When devices are added to a snapshot policy storage group, snapshot policies that apply to the storage group are applied to the added devices.
- When devices are removed from a snapshot policy storage group, snapshot policies that apply to the storage group are no longer applied to the removed devices.
- If overlapping snapshot policies are applied to storage groups, they run and take snapshots independently.

Unisphere provides compliance information for each snapshot policy that is directly associated with a storage group. Snapshot policy compliance is measured against the count and intervals of the existing snapshots. Snapshots must be valid (must still exist, must be in a non-failed state, and must be at the expected scheduled time). A snapshot could be missing due to it being manually terminated or due to a failure in the snapshot operation.

Snapshot compliance for a storage group is taken as the lowest compliance value for any of the snapshot policies that are directly associated with the storage group.

Compliance for a snapshot policy that is associated with a storage group is based on the number of valid snapshots within the retention count. The retention count is translated to a retention period for compliance calculation. The retention period is the snapshot interval multiplied by the snapshot maximum count. For example, a one hour interval with a 30 snapshot count means a 30-hour retention period.

The compliance threshold value for green to yellow is stored in the snapshot policy definition. Once the number of valid snapshots falls below this value, compliance turns yellow.

The compliance threshold value for yellow to red is stored in the snapshot policy definition. Once the number of valid snapshots falls below this value, compliance turns red.

In addition to performance level compliance, snapshot compliance is also calculated by polling the storage system once an hour for SnapVX related information for storage groups that have snapshot policies that are associated with them. The returned snapshot information is summarized into the required information for the database compliance entries.

When the maximum count of snapshots for a snapshot policy is changed, this changes the compliance for the storage group or service level combination. Compliance values are updated accordingly.

If a compliance calculation is performed during the creation of a snapshot, an establish-in-progress state may be detected. This is acceptable for the most recent snapshot but is considered failed for any older snapshot.

When a storage group and service level have only recently been associated and the full maximum count of snapshots has not yet been reached, Unisphere scales the calculation to the number of snapshots that are available and represents compliance accordingly until the full maximum count of snapshots has been reached. If a snapshot failed to be taken for a reason (such as the storage group or service level was suspended or a snapshot was manually terminated before the maximum snapshot count was reached), the compliance is reported as out of compliance appropriately.

When the service level interval is changed, the compliance window changes and the number of snapshots may not exist for correct compliance.

If a service level is suspended or a storage group or service level combination is suspended, snapshots are not created. Older snapshots fall outside the compliance window, and the maximum count of required snapshots is not found.

Manual termination of snapshots inside the compliance window results in the storage group or service level combination falling out of compliance.

Configuration of alerts related to snapshot policies is available from **Settings > Alerts > Alert Policies** on the Unisphere user interface.

NOTE: Snapshot policy offsets (the execution time within the RPO interval) and snapshot timestamps are both mapped to be relative to the clock (including time zone) of the local management host. If times are not synchronized across hosts, these appear different to users on those hosts. Even if they are synchronized, rounding that occurs during time conversion may result in the times being slightly different.

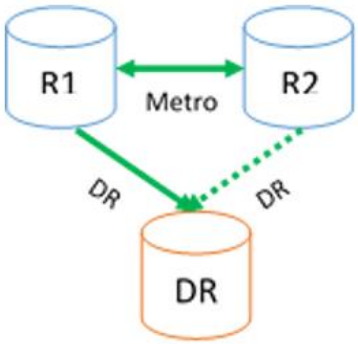
Unisphere supports the following snapshot policy management tasks:

- Create snapshot policies.
- View and modify snapshot policies.
- Associate a snapshot policy and a storage group with each other.
- Disassociate a snapshot policy and a storage group from each other.
- View snapshot policy compliance.
- Suspend or resume snapshot policies.
- Suspend or resume snapshot policies that are associated with one, more than one, or all storage groups.
- Set a snapshot policy snapshot to be persistent.
- Bulk terminate snapshots (not specific to snapshots associated with a snapshot policy).
- Delete snapshot policies.

Understanding SRDF/Metro Smart DR

SRDF/Metro Smart DR is a two-region high available (HA) disaster recovery (DR) solution. It integrates SRDF/Metro and SRDF/A enabling HA DR for a Metro session.

A session or environment name uniquely identifies each smart DR environment. It is composed of three arrays (MetroR1 array, MetroR2 array, DR array). All arrays contain the same number of devices and all device pairings form a triangle.



The MetroR1 array contains:

- One Metro SRDF Group that is configured to the MetroR2 array (MetroR1_Metro_RDFG)
- One DR SRDF Group that is configured to the DR array (MetroR1_DR_RDFG)
- Devices that are concurrent SRDF and are paired using MetroR1_Metro_RDFG and MetroR1_DR_RDFG.

The MetroR2 array contains:

- One Metro SRDF Group that is configured to the MetroR1 array (MetroR2_Metro_RDFG)
- One DR SRDF Group that is configured to the DR array (MetroR2_DR_RDFG).
- Devices that are concurrent SRDF and are paired using MetroR2_Metro_RDFG and MetroR2_DR_RDFG.

The DR array contains one DR SRDF Group that is configured to the MetroR1 array (DR_MetroR1_RDFG).

Unisphere for PowerMax supports the setup, monitoring, and management of a smart DR configuration using both UI and REST API.

Unisphere for PowerMax blocks attempts at using smart DR SRDF groups for other replication sessions, and also blocks certain active management on smart DR SRDF groups, including device expansion and adding new devices. This limitation can be overcome by temporarily deleting the Smart DR environment to perform these operations. Replication is never suspended so the Recovery Point Objective (RPO) is not affected.

Unisphere for PowerMax blocks attempts at SRDF active management of storage groups that are part of a smart DR environment.

Manage remote replication sessions

Unisphere supports the monitoring and management of SRDF replication on storage groups directly without having to map to a device group.

The SRDF dashboard provides a single place to monitor and manage SRDF sessions on a storage system, including device groups types R1, R2, and R21.

See the *Dell SRDF Introduction Guide* for an overview of SRDF.

Unisphere allows you to monitor and manage SRDF/Metro from the SRDF dashboard. SRDF/Metro delivers active/active high availability for non-stop data access and workload mobility – within a data center and across metro distance. It provides array clustering for storage systems enabling even more resiliency, agility, and data mobility. SRDF/Metro enables hosts and host clusters to directly access a LUN or storage group on the primary SRDF array and secondary SRDF array (sites A and B). This level of flexibility delivers the highest availability and best agility for rapidly changing business environments.

In a SRDF/Metro configuration, SRDF/Metro uses the SRDF link between the two sides of the SRDF device pair to ensure consistency of the data on the two sides. If the SRDF device pair becomes Not Ready (NR) on the SRDF link, SRDF/Metro must respond by choosing one side of the SRDF device pair to remain accessible to the hosts, while making the other side of the SRDF device pair inaccessible. There are two options which enable this choice: Bias and Witness.

The first option, Bias, is a function of the two storage systems taking part in the SRDF/Metro and is a required and integral component of the configuration. The second option, Witness, is an optional component of SRDF/Metro which allows a third storage system to act as an external arbitrator to avoid an inconsistent result in cases where the bias functionality alone may not result in continued host availability of a surviving nonbiased array.

Understanding SRDF groups

SRDF groups provide a collective data transfer path linking volumes of two separate storage systems. These communication and transfer paths are used to synchronize data between the R1 and R2 volume pairs that are associated with the SRDF group. At least one physical connection must exist between the two storage systems within the fabric topology.

These communication and transfer paths are used to synchronize data between the R1 and R2 volume pairs that are associated with the SRDF group. At least one physical connection must exist between the two storage systems within the fabric topology.

See the *Dell SRDF Introduction Guide* for an overview of SRDF.

The maximum number of supported SRDF groups differs by version:

Table 11. Maximum number of supported SRDF groups

OS	Maximum number of SRDF Groups supported			Group numbers
	per storage system	per director	per port	
5977 or 5978	250	250	250	1–250
10 (6079)	2000	2000	2000	1-2048

- **NOTE:** If both arrays are arrays running PowerMaxOS 10 (6079), up to 2000 SRDF groups can be defined across all the ports on a specific SRDF director or up to 2000 SRDF groups can be defined on one port on a specific SRDF director. A port on the array running PowerMaxOS 10 (6079) connected to an array running HYPERMAX OS 5977 or PowerMaxOS 5978 supports a maximum of 250 SRDF groups.
- When specifying a local or remote director for a storage system, you can select one or more SRDF ports.

SRDF group modes

SRDF groups provide a collective data transfer path linking volumes of two separate storage systems.

The following values can be set for SRDF groups:

Synchronous—This setting provides the host access to the source (R1) volume on a write operation only after the storage system containing the target (R2) volume acknowledges that it has received and checked the data.

Asynchronous—The storage system acknowledges all writes to the source (R1) volumes as if they were local volumes. Host writes accumulate on the source (R1) side until the cycle time is reached and are transferred to the target (R2) volume in one

delta set. Write operations to the target volume can be confirmed when the current SRDF/A cycle commits the data to disk by successfully de-staging it to the R2 storage volumes.

Semi Synchronous—The storage system containing the source (R1) volume informs the host of successful completion of the write operation when it receives the data. The SRDF (RA) director transfers each write to the target (R2) volume as the SRDF links become available. The storage system containing the target (R2) volume checks and acknowledges receipt of each write.

AC WP Mode On—(adaptive copy write pending) the storage system acknowledges all writes to the source (R1) volume as if it was a local volume. The new data accumulates in the cache until it is successfully written to the source (R1) volume and the remote director has transferred the write to the target (R2) volume.

AC Disk Mode On—This setting is used for situations requiring the transfer of large amounts of data without loss of performance. Use this mode to temporarily transfer the bulk of your data to target (R2) volumes, then switch to synchronous or semi-synchronous mode.

Domino Mode On—Ensures that the data on the source (R1) and target (R2) volumes are always synchronized. The storage system forces the source (R1) volume to a Not Ready state to the host whenever it detects one side in a remotely mirrored pair is unavailable.

Domino Mode Off—The remotely mirrored volume continues processing I/O operations per second with its host, even when an SRDF volume or link failure occurs.

AC Mode Off—Turns off the AC disk mode.

AC Change Skew—Modifies the adaptive copy skew threshold. When the skew threshold is exceeded, the remotely mirrored pair operates in the predetermined SRDF state (synchronous or semi-synchronous). When the number of invalid tracks drops below this value, the remotely mirrored pair reverts to the adaptive copy mode.

(R2 NR If Invalid) On—Sets the R2 device to Not Ready when there are invalid tracks.

(R2 NR If Invalid) Of—Turns off the (R2 NR_If_Invalid) On mode.

SRDF session modes

SRDF transparently remotely mirrors production or primary (source) site data to a secondary (target) site to users, applications, databases, and host processors.

Table 12. SRDF session modes

Mode	Description
Adaptive Copy	This mode allows the source (R1) volume and target (R2) volume to be out of synchronization by a number of I/O operations per second that are defined by a skew value.
Adaptive copy disk mode	Data is read from the disk, and the unit of transfer across the SRDF link is the entire track. While less global memory is consumed, it is typically slower to read data from disk than from global memory. Also, more bandwidth is used because the unit of transfer is the entire track. Also, because it is slower to read data from disk than global memory, device resynchronization time increases.
Adaptive Copy WP Mode	The unit of transfer across the SRDF link is the updated blocks rather than an entire track, resulting in more efficient use of SRDF link bandwidth. Data is read from global memory instead of disk, thus improving overall system performance. However, the global memory is temporarily consumed by the data until it is transferred across the link.
Synchronous	This mode provides the host access to the source (R1) volume on a write operation only after the storage system containing the target (R2) volume acknowledges that it has received and checked the data.
Asynchronous	The storage system acknowledges all writes to the source (R1) volumes as if they were local devices. Host writes accumulate on the source (R1) side until the cycle time is reached and are transferred to the target (R2) volume in one delta set. Write operations to the target device can

Table 12. SRDF session modes (continued)

Mode	Description
	be confirmed when the current SRDF/A cycle commits the data to disk by successfully de-staging it to the R2 storage volumes.
AC Skew	Adaptive Copy Skew - sets the number of tracks per volume the source volume can be ahead of the target volume. Values are 0–65535.

SRDF session options

SRDF transparently remotely mirrors production or primary (source) site data to a secondary (target) site to users, applications, databases, and host processors.

Table 13. SRDF session options

Session option	Description	Available with action
Bypass	This option bypasses the exclusive locks for the local or remote storage system during SRDF operations. Use this option only if you are sure that no other SRDF operation is in progress on the local or remote storage systems.	• Establish • Failback • Failover • Restore • Incremental Restore • Split • Suspend • Swap • Write Disable R1 • Ready R1 • Ready R2 • RWDisableR2 • Enable • Disable
Consistent	This option allows only a consistent transition from async to sync mode.	Activate
Consistency Exempt	This option allows you to add or remove volumes from an SRDF group that is in Async mode without requiring other volumes in the group to be suspended.	• Half Move • Move • Suspend
Establish	This option fails over the volume pairs, performs a dynamic swap, and incrementally establishes the pairs. This option is not supported when volumes operating in Asynchronous mode are read/write on the SRDF link. To perform a failover operation on such volumes, specify the Restore option.	• Failover
Force	This option overrides any restrictions and forces the operation, even though one or more paired volumes may not be in the expected state. Use caution when checking this option because improper use may result in data loss.	• Establish • Incremental Establish • Restore • Incremental Restore • Write Disable R1 • Ready R1 • Ready R2 • RWDisableR2 • Enable • Disable Swap

Table 13. SRDF session options (continued)

Session option	Description	Available with action
Immediate	This option causes the suspend, split, and failover actions on asynchronous volumes to happen immediately.	• Suspend • Split • Failover
NoWD	The No write disable option bypasses the check to ensure that the target of operation is write disabled to the host. This option applies to the source (R1) volumes when used with the Invalidate R1 option and to the target (R2) volumes when used with the Invalidate R2 option.	
SymForce	This option forces an operation on the volume pair including pairs that would be rejected. Use caution when checking this option because improper use may result in data loss.	• Restore • Incremental Restore • Write Disable R1 • Ready R1 • Ready R2 • RWDisableR2 • Enable • Disable Swap
Refresh R1	This option marks any changed tracks on the source (R1) volume to be refreshed from the target (R2) side.	• Swap
Refresh R2	This option marks any changed tracks on the target (R2) volume to be refreshed from the source (R1) side.	• Swap
Remote	This option is used when performing a restore or failback action with the concurrent link up, data that is copied from the R2 to the R1 is copied to the concurrent R2. These actions require this option.	• Restore • Incremental Restore • Failback
Restore	When the failover swap completes, invalid tracks on the new R2 side (formerly the R1 side) are restored to the new R1 side (formerly the R2 side). When used together with the Immediate option, the failover operation immediately deactivates the SRDF/A session without waiting two cycle switches for the session to terminate.	• Failover

SRDF/A control actions

SRDF transparently remotely mirrors production or primary (source) site data to a secondary (target) site to users, applications, databases, and host processors.

Table 14. SRDF/A control actions

Action	Activate Type	Write Pacing Type	Description
Activate	DSE	N/A	This type activates the SRDF/A Delta Set Extension feature. This feature extends the available cache space by using device SAVE pools.

Table 14. SRDF/A control actions (continued)

Action	Activate Type	Write Pacing Type	Description
	Write Pacing - this feature extends the availability of SRDF/A by preventing conditions that result in cache overflow on both the R1 and R2 sides.	Group write pacing	This type activates SRDF/A write pacing at the group level.
		Group and Volume Write Pacing	This type activates SRDF/A write pacing at the group level and the volume level.
		Volume Write Pacing	This type activates SRDF/A write pacing at the volume level.
	Write Pacing Exempt	N/A	This type activates write pacing exempt. Write pacing exempt allows you to remove a volume from write pacing.

SRDF group SRDF/A flags

SRDF groups provide a collective data transfer path linking volumes of two separate storage systems.

Table 15. SRDF group SRDF/A flag status

Flag	Status
(C) Consistency	X = Enabled; . = Disabled; - = N/A
(S) Status	A = Active; I = Inactive; - = N/A
(R) RDFA Mode	S = Single-session, M = MSC; - = N/A
(M) Msc Cleanup	C = MSC Cleanup required; - = N/A
(T) Transmit Idle	X = Enabled; = Disabled; - = N/A
(D) DSE Status	A = Active; I = Inactive; - = N/A
DSE (A) Autostart	X = Enabled; = Disabled; - = N/A

Understanding migration

Unisphere supports a migration application to provide a method for migrating data from a source storage system to a target storage system.

Migration can be defined as a one-way movement of an application or applications from a source storage system to a target storage system with the intention of retiring the source.

PowerMax Data Mobility can be defined as the ongoing movement of data of an application between storage systems for various reasons: load balancing, performance optimization, HA, and migration. Data Mobility incorporates Non-Disruptive Migration (NDM), Minimally Disruptive Migration (MDM), and Open Minimally disruptive migration (O-MDM).

NDM provides a method for migrating data from a source storage system to a target storage system without application host downtime. NDM enables you to migrate storage group (application) data (the storage groups must have masking views) in a non-disruptive manner with no downtime.

MDM enables migrations on the same supported platforms as non-disruptive migration it but requires a short application outage. The outage is because that the non-disruptive nature of migration is heavily dependent on the behavior of multi-pathing software to detect or enable or disable paths and is not in the control of Dell (except for PowerPath).

NDM provides non-disruptive and minimally disruptive support from VMAX3, VMAX All Flash, and PowerMax arrays running 5977 and 5978 firmware to PowerMax 2500/8500.

O-MDM provides integration of Open Replicator into Data Mobility allowing automation of migrations. Provides migration from any VMAX array (5876, 5977, 5978), Unity, XtremIO, or any FC connected array to a PowerMax 2500 or PowerMax 8500 storage system.

Metro-based NDM is built upon SRDF/Metro active/active technology with the Metro instant activate (Metro-IA) feature. Benefits of using Metro-based NDM include the following:

- Allows the complete migration from a VMAX3 or VMAX All Flash arrays to VMAX All Flash or PowerMax arrays without the need for downtime.
- Leverages Metro technology to reduce the required commands users must issue by removing the need to cut over.
- Manageable by familiar and simple user interfaces: Solutions Enabler and Unisphere for PowerMax.
- Allows migrations to be canceled and failed back to the source array for any reason before commit.
- Allows users to precopy all or a large portion of data before bringing the host live to the new array. The precopy reduces the impact to the application during the migration period.

More migration information is available in the *Solutions Enabler Array Controls and Management CLI User Guide* and the *Non-Disruptive Migration Best Practices and Operational Guide*. Source side service levels are automatically mapped to target side service levels.

Non-Disruptive Migration applies to open systems or FBA devices only.

Non-Disruptive Migration supports the ability to reduce data on all-flash storage systems while migrating.

A Non-Disruptive Migration session can be created on a storage group containing session target volumes (R2s) where the SRDF mode is synchronous. The target volumes of a Non-Disruptive Migration session may also have a SRDF/Synchronous session that is added after the Non-Disruptive Migration session is in the cutover sync state.

Suggested best practices

- Try to migrate during slow processing times; QoS can be used to throttle copy rate.
- Use more SRDF links, if possible, to minimize impact:
  - Two is the minimum number of SRDF links allowed; Non-Disruptive Migration can use up to eight SRDF links.
  - More links = more IOPS, lower response time.
- Use dedicated links as they yield more predictable performance than shared links.

You can migrate masked storage groups where the devices can also be in other storage groups. Examples of overlapping storage devices include:

- Storage groups with the exact same devices, for example, SG-A has devices X, Y, Z; SG-B has devices X, Y, Z.
- Devices that overlap, for example, SG-A has devices X, Y, Z ; SG-B has devices X, Y.
- Storage groups where there is overlap with one other migrated SG, for example, SG-A has devices X, Y, Z ; SG-B has devices W, X, Y; SG-C has devices U, V, W.

The following migration tasks can be performed from Unisphere:

- Setting up a migration environment—Configures source and target storage system infrastructure for the migration process.
- Viewing migration environments
- Creating a Migration session—Duplicates the application storage environment from the source storage system to the target storage system.
- Viewing Migration sessions
- Viewing Migration session details
- Cutting over a Migration session—Switches the application data access form the source storage system to the target storage system and duplicates the application data on the source storage system to the target storage system.
- Optional: Stop synchronizing data after Migration cutover and Start synchronizing data after Migration cutover—stop or start the synchronization of writes to the target storage system back to the source array. When stopped, the application runs on the target storage system only.
- Canceling a Migration session—cancels a migration that has not yet been committed.
- Committing a Migration session—Removes application resources from the source storage system and releases the resources that are used for migration. The application permanently runs on the target array.
- Recovering a Migration session—Recovers a migration process following an error.
- Removing a migration environment—Removes the migration infrastructure.

Understanding Virtual Witness

The Virtual Witness feature supports a third party that the two storage systems consult when they lose connectivity with each other, that is, their SRDF links go out of service.

When SRDF links go out of service, the Witness helps to determine, for each SRDF/Metro Session, which of the storage systems should remain active (volumes continue to be read and write to hosts) and which goes inactive (volumes not accessible).

For additional information about vWitness, see the *Dell SRDF/Metro vWitness Configuration Guide*.

The following vWitness tasks can be performed from Unisphere.

- Adding a Virtual Witness
- Viewing Virtual Witness instances
- Viewing Virtual Witness instance details
- Enabling a Virtual Witness
- Disabling a Virtual Witness
- Removing a Virtual Witness

Understanding Open Replicator

Open Replicator is a non-disruptive migration and data mobility application.

When the Open Replicator control volumes are on a storage system, the following session options cannot be used:

- Push
- Differential
- Precopy

There are many rules and limitations for running Open Replicator sessions. See the *Solutions Enabler Migration CLI Product Guide* before creating a session. For a quick reference, see Open Replicator session options.

Open Replicator session options

Open Replicator is a non-disruptive migration and data mobility application.

Depending on the operation that you are performing, some of the following options may not apply.

Table 16. Open Replicator session options

Session Option	Used with UI operation	Description
Consistent	Activate	This option causes the volume pairs to be consistently activated.
	Donor Update Off	Consistently stops the donor update portion of a session and maintains the consistency of data on the remote volumes.
Copy	Create	Volume copy takes place in the background. This behavior is the default for both pull and push sessions.
Cold	Create	Control volume is write-disabled to the host while the copy operation is in progress. A cold copy session can be created provided one or more directors discovers the remote device.
Differential	Create	This option creates a one-time full volume copy. Only sessions that are created with the differential option can be re-created. For push operations, this

Table 16. Open Replicator session options (continued)

Session Option	Used with UI operation	Description
		option is selected by default. For pull operations, this option is cleared by default (no differential session).
Donor Update	Create	This option causes data that is written to the control volume during a hot pull to also be written to the remote volume.
	Incremental Restore	Maintains a remote copy of any newly written data while the Open Replicator session is restoring.
Force	• Terminate • Restore	Select the **Force** option if the copy session is in progress. This option allows the session to continue to copy in its current mode without donor update.
	Donor Update Off	Select the **Force** option if the copy session is in progress. This option allows the session to continue to copy in its current mode without donor update.
Force Copy	Activate	This option overrides any volume restrictions and allows a data copy. For a push operation, remote capacity must be equal to or larger than the control volume extents and conversely for a pull operation. The exception to this rule is when you have pushed data to a remote volume that is larger than the control volume, and you want to pull the data back, you can use the **Force_Copy** option.
Front-End Zero Detection	Create	This option enables front end zero detection for thin control volumes in the session. Front-end zero detection looks for incoming zero patterns from the remote volume, and instead of writing the incoming data of all zeros to the thin control volume, the group on the thin volume is de-allocated.
Hot	Create	Hot copying allows the control device to be read/write online to the host while the copy operation is in progress. All directors that have the local devices mapped are required to participate in the session. A hot copy session cannot be created unless all directors can discover the remote device.
Nocopy	Activate	Temporarily stops the background copying for a session by changing the state to CopyOnAccess or CopyOnWrite from CopyInProg.
Pull	Create	A pull operation copies data to the control device from the remote device.
Push	Create	A push operation copies data from the control volume to the remote volume.
Precopy	• Create • Recreate	For hot push sessions only, begins immediately copying data in the

Table 16. Open Replicator session options (continued)

Session Option	Used with UI operation	Description
		background before the session is activated.
SymForce	Terminate	Forces an operation on the volume pair including pairs that would be rejected. Use caution when checking this option because improper use may result in data loss.

Understanding device groups

A device group is a user-defined group that consists of devices that belong to a locally attached array. Control operations can be performed on the group as a whole, or on the individual device pairs in the group. By default, a device can belong to more than one device group.

The user can create a legacy TF emulation from source devices with a SNAP VX snapshot. The prerequisites are:

- A SNAP VX storage group with a snapshot must exist.
- A device group must already have been created from this storage group.
- The device group must also have enough candidate target devices to create the required TF emulation session.

Understanding TimeFinder/Mirror sessions

TimeFinder/Mirror is a business continuity solution that enables the use of special business continuance volume (BCV) devices. Copies of data from a standard device (which are online for regular I/O operations from the host) are sent and stored on BCV devices to mirror the primary data. Uses for the BCV copies can include backup, restore, decision support, and applications testing. Each BCV device has its own host address, and is configured as a stand-alone device.

- On storage systems running HYPERMAX OS 5977 or PowerMaxOS 5978, TimeFinder/Mirror operations are mapped to their TimeFinder/SnapVX equivalents.
- TimeFinder operations are not supported on Open Replicator control volumes on storage systems running HYPERMAX OS 5977 or PowerMaxOS 5978.

The TimeFinder/Mirror dashboard provides a single place to monitor and manage TimeFinder/Mirror sessions on a storage system.

Understanding TimeFinder SnapVX

TimeFinder SnapVX is a local replication solution that is designed to nondisruptively create point-in-time copies (snapshots) of critical data.

TimeFinder SnapVX creates snapshots by storing changed tracks (deltas) directly in the Storage Resource Pool of the source volume. With TimeFinder SnapVX, you do not need to specify a target volume and source/target pairs when you create a snapshot. If the application ever needs to use the point-in-time data, you can create links from the snapshot to one or more target volumes. If there are multiple snapshots and the application needs to find a particular point-in-time copy for host access, you can link and relink until the correct snapshot is located.

Understanding Performance Management

Using the Unisphere Performance Management application, you can gather, view, and analyze performance data to troubleshoot and optimize the storage systems.

Performance Management covers the following areas:

- Dashboards - Display predefined, user-defined custom dashboards, and templates.
- Charts - Create custom charts across multiple categories, metrics, time, or intervals.
- Analyze - Provide in-depth analysis on storage system data for various collection ranges.

- Heatmap - Display hardware instances that are represented as colored squares, with the colors indicating utilization levels.
- Reports - Create, manage, and run performance reports.
- Real-Time Traces - Create, manage, and run performance real-time traces .
- Databases - Manage Performance database tasks, for example, back up, restore, delete, and also individual performance database information.
- Plan - Provide performance projection capacity dashboards displaying predicted future data that is based on linear projection.

Suggesting intelligent thresholds

The Unisphere Performance Management application allows the user to gather, view, and analyze performance data to troubleshoot and optimize the storage systems.

About this task

You can use the intelligent threshold functionality to analyze historical trends of performance metrics and system capability, and use the data to help to set appropriate thresholds that highlight real issues.

To use this functionality, the storage system must be registered for the collection of performance data, you need a minimum of 15 days of performance data. Only the **System** and **Storage Group** categories of metrics are supported.

To create performance threshold alerts:

Steps

1. Select  .
2. Select **Alerts** > **Performance Thresholds and Alerts**.
3. Click **Suggest Intelligent Thresholds**.
   The following properties are displayed in the **Suggest Intelligent Thresholds** dialog:
   - **System**—The storage system
   - **Category**—The category to which the selected metric belongs. Only the **System** and **Storage Group** categories are supported for intelligent thresholds.
   - **KPI/All**—Filters the list of metrics by whether they are KPIs or not.
   - **Metric**—The metric for which you want to generate intelligent thresholds.
   - Intelligent Threshold Alert chart—Displays the suggested threshold values and projection data.
   - **Close**—Closes the dialog and cancels the operation.
   - **Apply**—Generates suggested threshold values.
4. Select **All** or a particular storage **System**.
5. Select the **Category**, **Instance** (for **Storage Group** categories), and **Metric** that you want to use as a template when configuring intelligent thresholds. You can select **KPI** or **All** to filter the metrics displayed.
6. Click **Suggest Thresholds**.
   The Intelligent Threshold Alert charts are generated. The upper chart displays suggested values for the First Threshold. The lower chart displays suggested values for the optional Second Threshold.
7. To create threshold alerts with the suggested values, click **Apply**.
   The **Create Performance Threshold and Alert** dialog is displayed, with the metric and suggested threshold values populated.
8. If required, edit the suggested threshold and alert values.
9. To create the threshold for the given instances and metric, click **OK**.

Understanding Unisphere support for VMware

Unisphere supports the discovery of vCenters or ESXi servers (using a read-only user) and integrates the information into Unisphere. VMware information is connected to its storage extents, and this information enables investigation of any storage-related issues.

Unisphere support for VMware provides the storage admin access to all the relevant storage-related objects to an ESXi server and also helps troubleshooting storage performance-related issues to the ESXi server.

You can, as a read-only user, discover at the vCenter level and discover an individual ESXi server. If a vCenter is discovered, all ESXi servers under that vCenter are discovered.

Once the user adds VMware information, all other users of Unisphere can access this information.

The minimum version number that vCenter supports is version 5.5.

Understanding eNAS

Embedded NAS (eNAS) integrates the file-based storage capabilities of VNX arrays into storage systems running HYPERMAX OS 5977 or PowerMaxOS 5978.

With this integrated storage solution, the Unisphere StorageAdmin provisions storage to eNAS data movers, which trigger the creation of storage pools in VNX. Then, users of Unisphere for VNX can use the storage pools for file-level provisioning, for example, creating file systems, file shares.

Unisphere provides the following features to support eNAS:

File System dashboard	Provides a central location from which to monitor and manage integrated VNX file services.
Provision Storage for File wizard	Allows you to provision storage to eNAS data movers.
Launch Unisphere for VNX	Allows you to link and launch Unisphere for VNX.

Understanding iSCSI

Unisphere provides monitoring and management for Internet Small Computer Systems Interface (iSCSI) directors, iSCSI ports, iSCSI endpoints, IP interfaces, and IP routes.

iSCSI is a protocol that uses the TCP to transport SCSI commands, enabling the use of the existing TCP/IP networking infrastructure as a SAN. As with SCSI over Fibre Channel (FC), iSCSI presents SCSI endpoints and devices to iSCSI initiators (requesters). Unlike NAS, which presents devices at the file level, iSCSI makes block devices available from the network. Block devices are presented across an IP network to your local system, and can be consumed in the same way as any other block storage device.

The iSCSI changes address the market needs that originate from the cloud or service provider space, where a slice of infrastructure, for example, computes, network and storage, is assigned to different users (tenants). Control and isolation of resources in this environment is achieved by the iSCSI changes. Also, more traditional IT enterprise environments also benefit from this new functionality. The changes also provide greater scalability and security.

Understanding NVMe/TCP

Unisphere provides monitoring and management for NVMe/TCP directors, NVMe/TCP ports, NVMe/TCP endpoints, IP interfaces, and IP routes.

NVMe/TCP is a NVME-oF technology that supports high performance with lower deployment costs and reduced design complexity. NVMe/TCP defines a new storage networking fabric for the NVMe block storage protocol over a storage networking fabric, and provides the capability of extending NVMe across the entire data center .

The Unisphere UI supports the following operations (on storage systems running PowerMaxOS 10 (6079)):

- View the NVMe dashboard.
- Configure NVMe/TCP using a wizard.
- View NVMe/TCP directors.
- View and configure NVMe/TCP endpoints.
- View NVMe/TCP ports.
- View and configure IP routes.
- View and configure IP interfaces.

Understanding PowerMax File for storage systems

Network-attached storage is a file-level storage architecture that makes stored data more accessible to networked devices.

A storage system running PowerMaxOS 10 (6079) with embedded Unisphere supports software-defined network-attached storage (PowerMax File).

PowerMax File support a file services architecture that provides a reliable, high performance, highly available, and highly scalable system. PowerMax File runs as a container instance inside each file guest offering based on the customer configuration.

Unisphere in the Embedded Element Manger (EEM) manages a single instance of PowerMax File that is present locally. The Unisphere that is installed on an external host does not manage any PowerMax File instances.

PowerMax File uses virtualized Network-Attached Storage (NAS) servers that use the SMB, NFS, and FTP protocols to catalog, organize, and transfer files within file system shares and exports.

A NAS server, the basis for multi-tenancy, must be created before you can create file-level storage resources. NAS servers are responsible for the configuration parameters on the set of file systems that it serves.

Network File System (NFS) is an access protocol that enables users to access files and folders on a network. You can create an NFS export to make file system paths on your storage system available for mounting by NFS clients.

Server Message Block (SMB) is an access protocol that allows remote file data access from clients to hosts on a network. An SMB share, also known as an SMB file share, is a shared resource on an SMB server.

VLAN and Jumbo frames are supported for File services.

Services such as anti-virus, scheduled snapshots, and Network Data Management Protocol (NDMP) backups ensure that the data on the file systems is well protected.

Common Event Enablers (CEE) is an externally deployable software from Dell (primarily on Windows). This software has two packages - Common anti-virus Agent (CAVA) and Common Event Publishing Agent (CEPA). CAVA provides an anti-virus solution to clients using a NAS server. It uses an industry-standard SMB protocol in a Windows Server environment. CAVA uses third-party anti-virus software to identify and eliminate known viruses before they infect files on the storage system. The CEE CEPA ecosystem consists of dozens of applications that are designed to process SMB and NFS file and directory event notifications.

The high-level sequence of tasks to configure PowerMax File for storage systems running PowerMaxOS 10 (6079) is as follows:

1. Configure subnets (subnet configuration is required in order to create a NAS server).
2. Create a NAS server (a NAS server is required in order to create a file system).
3. Create File Systems (file systems enable you to partition data for your users).
4. Protect File Systems (optionally protect file system data using snapshots or replication).

The detailed sequence of tasks to configure PowerMax File is as follows:

1. View node inventory (under **System > File Configuration> Node Inventory** tab).
2. Configure subnets. Create and configure bond devices (under **System > File Configuration> Network Devices** tab).
3. Create and configure NAS servers (under **Storage > File**). Other properties and controls that are associated with NAS servers are in the following tabs:
   - **Details**
   - **Network > File Interface**
   - **Network > Routes to external services**
   - **Sharing Protocols > SMB Server**
   - **Sharing Protocols > NFS Server**
   - **Sharing Protocols > FTP**
   - **Sharing Protocols > User Mapping**
   - **Naming services > DNS**
   - **Naming services > UDS**
   - **Naming services > Local Files**
   - **Security > Antivirus**
   - **Security > Kerberos**
   - **Data protection > Snapshot policy**
   - **Data protection > Replication**
   - **Backup & Events > NDMP**—You can configure standard backup for the NAS servers using Network Data Management Protocol (NDMP). NDMP provides a standard for backing up file servers on a network.
   - **Backup & Events > DHSM**—Distributed Hierarchical Storage Management (DHSM) supports file-based archiving.

- **Backup & Events > Events pool**
- **Nodes**

4. Create and configure file systems (under **Storage > File**). Other properties and controls that are associated with file systems are in the following tabs :
   - **Details**
   - **Snapshots**—Snapshots can be used for restoring individual files or the entire file system back to a previous point in time.
   - **User Quotas**—Quota management allows you to place limits on the amount of space that can be consumed in order to regulate file system storage consumption. User quotas are set at a file system level and limit the amount of space a user may consume on a file system.
   - **Tree Quotas**—Quota management allows you to place limits on the amount of space that can be consumed in order to regulate file system storage consumption. Quota trees limit the maximum size of a directory on a file system. Unlike user quotas, which are applied and tracked on a user-by-user basis, quota trees are applied to directories within the file system.
   - **File Level Retention**—When a snapshot is created, it can be configured to have no automatic deletion or retention until a specific date and time. If retention is set, the snapshot is automatically deleted upon reaching the retention date. This does not prevent the snapshot from being deleted before the retention date.
5. Create and configure SMB Shares (under **Storage > File > SMB Shares**).
6. Create and configure NFS Exports (under **Storage > File > NFS exports**). You can create an NFS export to make file system paths on your storage system available for mounting by NFS clients.
7. Create Global Namespaces (under **Storage > File > Global Namespaces**).
8. Create and configure snapshot policies (under **Data Protection > File Protection> Snapshot policies**).
9. Create and configure replication (under **Data Protection > File Protection > Replication**. Properties and controls that are associated with replication are:
   - **File control network**
   - **Remote connection**
   - **Replication sessions**

Migration support is provided for file objects, system configuration, and user data from eNAS to PowerMax File.

The sequence of tasks to migrate from eNAS to PowerMax File is as follows:

1. Set up File migration network for data import (under **System > File Migration > Network** tab).
2. Set up File migration management connection to the remote storage system (under **System > File Migration > Remote Systems** tab).
3. Set up File migration sessions by importing sessions from the remote source NAS server (under **Data Protection > File Migration > File Imports**).

The state transitions for File migration are outlined in the following diagram and tables.

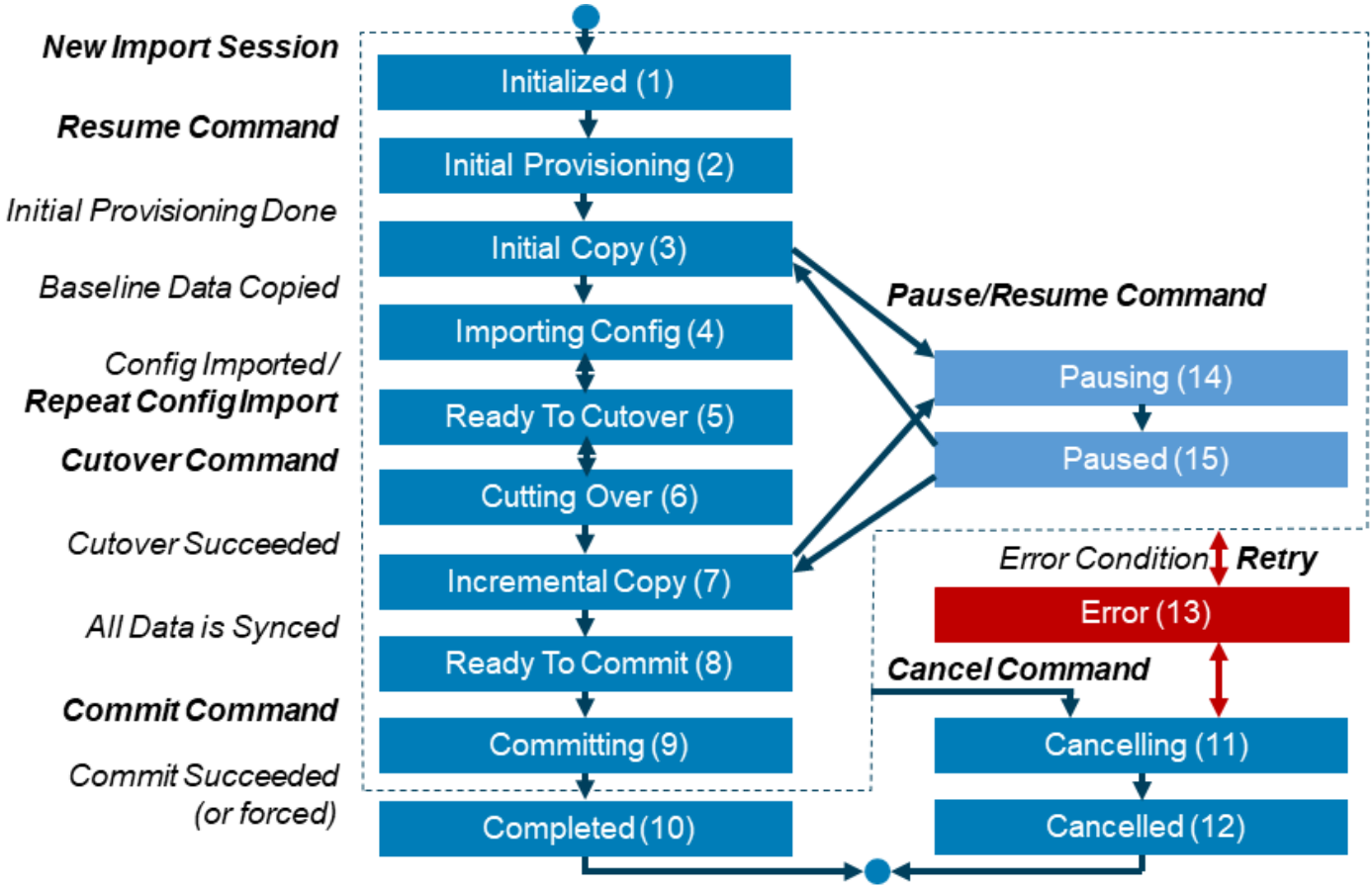


Table 17. State transitions for File migration

State	Name	Resume	Pause	Cutover	Commit	Cancel	Modify
1	Initialized	Allowed	-	-	-	Allowed	Allowed
2	Initial_Provisioning	-	-	-	-	Allowed	Allowed
3	Initial_Copy	-	Allowed	-	-	Allowed	-
4	Importing_Config	-	-	-	-	Allowed	Allowed
5	Ready_to_Cutover	Allowed*	-	Allowed	-	Allowed	Allowed
6	Cutting_Over	-	-	-	-	Allowed	-
7	Incremental_Copy	-	Allowed	-	-	Allowed	-
8	Ready_to_Commit	-	-	-	Allowed	Allowed	-
9	Committing	-	-	-	-	-	-
10	Completed	-	-	-	-	-	-
11	Cancelling	-	-	-	-	-	-
12	Canceled	-	-	-	-	-	-
13	Error	Allowed**	Allowed**	Allowed**	Allowed**	Allowed**	Allowed**
14	Pausing	-	-	-	-	Allowed	-
15	Paused	Allowed	-	-	-	Allowed	-

* If repeatConfigImport is true.

** See the following table

Table 18. State transitions for File migration Error state

State ID	State Name	Stage	Name	Resume	Pause	Cutover	Commit	Cancel	Modify
13	Error	1	Initialized	Allowed	-	-	-	Allowed	Allowed
-	-	2	Initial_Provisioning	Allowed	-	-	-	Allowed	Allowed
-	-	3	Initial_Copy	Allowed	Allowed	-	-	Allowed	-
-	-	4	Importing_Config	Allowed	-	-	-	Allowed	Allowed
-	-	5	Ready_to_Cutover	Allowed*	-	Allowed	-	Allowed	Allowed
-	-	6	Cutting_Over	-	-	Allowed	-	Allowed	-
-	-	7	Incremental_Copy	Allowed	Allowed	-	-	Allowed	-
-	-	8	Ready_to_Commit	-	-	-	Allowed	Allowed	-
-	-	9	Committing	-	-	-	Allowed	-	-
-	-	10	Completed	-	-	-	-	-	-
-	-	11	Cancelling	-	-	-	-	Allowed	-
-	-	12	Canceled	-	-	-	-	-	-

* If repeatConfigImport is true.

Understanding serviceability

Unisphere provides a serviceability application that supports the deployment of update packages for PowerMax OS and embedded applications.

The Unisphere for PowerMax Serviceability application integrates functionality that was previously supported by the stand-alone vApp manager application. The functionality enables the user to:

- View the version information and status of the PowerMax OS and embedded applications.
- Configure an NTP server.
- Modify external NAT IP addresses.
- Download logs that log the changes that the Serviceability applications make.
- Configure Solutions Enabler base and service settings.
- Set **Symavoid** on both instances on an embedded system.
- Control service access.
- Perform operations on VASA container.
- Configure VASA Provider settings for VASA containers.
- Manage certificates for SE/vWitness.
- Configure nethost entries.
- Manage certificates for Unisphere.
- Download performance reports, performance databases, and system databases.

Application -> VASA

- Provides and certifies VASA 5.0 support on the PowerMax 2500 and 8500 storage systems.
- Provides the ability to non-disruptively upgrade a VASA provider, running on a PowerMax 2500 or 8500 storage system, to a version that supports VASA 5.0. VASA Provider, when updated to version 5.0, requires VASA Provider to be unregistered and registered back to vCenter to build up the access control for the vCenter by running the registration steps again. This step is required since the VASA Provider would be building a database of trusted clients from scratch. VASA Provider expects that all provisioning operations arrive from the virtual host.
- Once a storage container is provisioned to a VASA 5.0 supported client, earlier clients (< VASA 5) are unable to access this storage container.
- In a Multi vCenter environment, it is recommended to unregister the standby VASA Provider first. Expect VASA Provider failover from vCenter if the active VASA Provider is unregistered first.

- In a Multi vCenter setup, if the same storage container is shared across multiple vCenters, then once the VASA Provider is upgraded to 10.2.0, it is required that all vCenters in the data center be upgraded to the VASA 5.0 supported vSphere version in order to allow access to this storage container.
- After the VASA Provider upgrade to 10.2.0, for earlier client versions (< VASA 5.0) and VMCA certificate management, the VP_RETAIN_CERTIFICATE flag should be set to false. See the *Unisphere for PowerMax Release Notes* for additional details.
- Ensure that you register the active VASA first if registering on multiple vCenters.

Understanding PowerMax software system profiles and compliance

A PowerMax software system profile is a subset of system settings that are associated with at least one storage system (array). The settings are defined once as part of a system profile, and these settings can be applied to one or more storage systems.

This feature supports the creation of a software profile that can be applied to one or more storage systems running PowerMaxOS 10 (6079) that are local to the managing Unisphere for PowerMax. The profile covers several alert and performance settings. The compliance part of the feature relates to monitoring of changes (that occur within a certain user-specified time window) to any local storage system (running PowerMaxOS 10 (6079)) that is part of a profile.

The Unisphere UI supports the following operations (on storage systems running PowerMaxOS 10 (6079)):

- View the system profiles dashboard.
- Create, modify, and delete system profiles.
- Apply a system profile.
- Add a storage system to a profile.
- View compliance status.
- Create, modify, and delete change control window.
- View audit log records for changes outside the change control window.
- Import, export, and clone a change control window.

View physical view

Unisphere supports the display of the physical view for a storage system.

Prerequisites

See Roles and associated permissions to determine the user roles that can perform this task.

About this task

A physical view is supported for storage systems running PowerMaxOS 10 (6079). The view displays how the hardware is comprised along with its associated components. The view consists of the front or rear bay images, list of directors and DAEs within the system. Hovering over each component displays a tooltip detailing the name of the component.

Steps

1. Select the storage system.
2. Select **System** > **Physical View**.
3. Click the **Front** radio button.

   The front physical view displays. Hovering over a component within the physical view results in the display of a tooltip displaying the FRU ID.

4. Click the **Rear** radio button.

   The rear physical view displays. Hovering over a component within the physical view results in the display of a tooltip displaying the FRU ID.

5. Depending on the storage system selected, click the refresh icon or click the **Refresh Physical View** to retrieve and display the most recent physical view details.
6. To view an animation of the steps involved in a component procedure, select an operation (replace, install, uninstall, remove, add, open, or close) from the **Animations** drop-down menu.

7. From the rear physical view, click the **Cabling** checkbox and select one of the options on the **Cabling** drop-down menu to view the related cable connection details.

Replace faulted SFP

Unisphere for PowerMax supports a small form-factor pluggable component (SFP) replacement policy. This policy specifies whether it is the customer or Customer Support that is responsible for the replacement of SFPs.

Prerequisites

See Roles and associated permissions to determine the user roles that can perform this task.

This task applies to storage systems running PowerMaxOS 10.2.0 that support a customer replaceable small form factor pluggable component (SFP) replacement policy.

The SFP replacement procedure occurs in response to an automatic SFP part and email dispatch.

About this task

NOTE: If the SFP replacement is not successful or if a user attempted stop of an in-progress replacement occurs, the lock in SymmWin is kept by the replacement process. The result of this lock is that other SymmWin processes cannot be run. You must contact Customer Support to resolve this issue.

Steps

1. Select the storage system.
2. Select **System** > **Physical View**.
3. Click **Replace Faulted SFPs**
4. Select the faulted SFP to be replaced.
5. Click **Run Now**
6. Read the warning concerning SFP replacement and click **OK** to proceed.
7. On the **CRU Replacement Instructions** dialog, perform the tasks that are listed in the **Replacement Instructions** section.

   If you decide that you do not want to proceed with the replacement, click **Stop Replacement**. You can go to the Job List View to view the progress of the SFP replacement task and also stop an in-progress replacement.
8. Select the slider next to the **Export** button to zoom in and out on the hardware view.
9. Click **Export** to export the contents of the dialog to a .jpg file.
10. Go to the Job List View to monitor the status of the SFP replacement task.
11. When the Job List View gets updated to indicate that the SFP replacement task has been successful, go back to the **Replace Faulted SFPs** dialog and click **Mark as Completed**.

    You can also click **Mark as Completed** from the Job List View after it gets updated to indicate that the SFP replacement task has been successful.

    NOTE:

    If the SFP replacement is not successful or if a user-attempted stop of an in-progress replacement occurs, the lock in SymmWin is kept by the replacement process. The result of this lock is that other SymmWin processes cannot be run. You must contact Customer Support to resolve this issue.

View FPIN information

The SAN Health Report view and the Fabric Performance Impact Notifications (FPINs) view allow the user to determine the health of their storage network.

Prerequisites

See Roles and associated permissions to determine the user roles that can perform this task.

About this task

Fabric Performance Impact Notification (FPIN) is a facility that is provided by switch vendors. The switch reports to the storage system using the following types of notifications:

- Link Integrity Notification
- Path Congestion Notification
- Peer-path Congestion Notification (Switch congestion that may spread to this path).
- Delivery Notification

Fabric alerts are received from the switch and forwarded for display in the alerts view. These alerts allow performance degradation analysis to be performed more quickly.

You can view FPIN alert information (only available on storage systems running PowerMaxOS 10.1.0 or later) in the Alerts view. There is an option in the Alerts Policies dialog to disable or enable FPIN Alerts (enabled by default). Disabling FPIN alerts results in new FPIN alerts no longer being displayed in the Alerts view.

Steps

1. Select a storage system running PowerMaxOS 10.1.0 or later.
2. Select the **System Health** tab.
3. In the **Actions** panel, click **View SAN Health**.

   A screen is displayed with two tabs: a **SAN Health Report** tab and a **FPINs** tab. The following properties display in the **FPINs** tab:

   - **Dir:Port** —Dir:Port ID
   - **Detect WWN**—Detect WWN
   - **Protocol**—Protocol, for example, SCSI FC
   - **Attach WWN**—Attach WWN
   - **FCID**—Fibre Channel ID
   - **Detection Time**—The time that the notification was detected.
   - **Storage Groups**—Number of storage groups impacted by the FPIN

4. Select a FPIN and click ⓘ to view additional properties.

   The following properties display for a FPIN:

   - **FPIN Type**—FPIN type
   - **Director**—Director ID
   - **Port**—Port associated with the director
   - **Protocol**—Protocol, for example, SCSI FC
   - **Attach WWN**—Attach WWN
   - **Detect WWN**—Detect WWN
   - **FCID**—Fibre Channel ID
   - **Detection Time**—The time that the notification was detected.
   - **Storage Groups**—Number of storage groups affected by the FPIN. Click the number next to **Storage Groups** to view the list of these storage groups.

   The following properties display for a selected storage group:

   - **Director Identification**—Director ID
   - **Director Port**—Port associated with the director
   - **Array Negotiated Speed (Gb/Sec)**—Array negotiated speed
   - **Director Port Speed (Gb/Sec)**—Director port speed
   - **Director Identification**—Director ID
   - **Director Port**—Director port
   - **Array Negotiated Speed (Gb/Sec)**—Array negotiated speed
   - **Director Port Speed (Gb/Sec)**—Director port speed

View CloudIQ feed

Unisphere can send data to CloudIQ that is related to configuration, performance, health scores, alerts, and security settings. CloudIQ can analyze this data and where it detects issues, it can generate recommendations for remediation. These recommendations are sent to Unisphere.

Prerequisites

See Roles and associated permissions to determine the user roles that can perform this task.

Steps

1. Select **EVENTS** > **CloudIQ Feed** to open the **CloudIQ Feed** view.

   The following properties display:

   - **Deduction**—Deduction ID
   - **Object**—Component to which the deduction is related.
   - **Last Updated**—Date or time of last update
   - **Description**—Description of the deduction

2. Optional: Select a deduction and click ⓘ to view more details and review recommendations.

Review recommendations and take action

Unisphere can send data that is related to configuration, performance, health scores, alerts, and security settings to CloudIQ. CloudIQ can analyze this data and when it detects issues, it can generate recommendations for remediation, which are sent to Unisphere.

Prerequisites

See Roles and associated permissions to determine the user roles that can perform this task.

Steps

1. Select the storage system.
2. Select **EVENTS** > **CloudIQ Feed** to open the **CloudIQ Feed** view.
3. Select a deduction and click ⓘ to view more details.
4. Click **Review**.

   A dialog is displayed specifying a recommendation and one or more potential actions.
5. Click the checkbox next to your chosen action.
6. Select the storage system.
7. On the **Summary** page, review the details and do one of the following:
   - Click **Run Now** to perform the operation now.
   - Expand **Run Now** and click **Add to Job List** to add this task to the job list, from which you can schedule or run the task at your convenience.

Unisphere for PowerMax localized for Japanese market

The Unisphere for PowerMax user interface and online help have been translated into Japanese.

However, strings in the following products are not translated:

- Solutions Enabler - This includes any Symm component names that are created using symapi. These component names should not accept Japanese/non-ASCII characters.
- Unisphere for PowerMax REST API
- GuestOS
- Embedded Unisphere including PowerMax File

- Unisphere for PowerMax Performance Viewer
- Third party strings

Where to get help

This topic outlines where to access information about Unisphere for PowerMax.

The Dell Technologies Support site contains important information about products and services including drivers, installation packages, product documentation, knowledge base articles, and advisories.

A valid support contract and account might be required to access all the available information about a specific Dell Technologies product or service.

Your comments

Your suggestions help continue to improve the accuracy, organization, and overall quality of the user publications. Send your feedback to content feedback.

Notes, cautions, and warnings

NOTE: A NOTE indicates important information that helps you make better use of your product.

CAUTION: A CAUTION indicates either potential damage to hardware or loss of data and tells you how to avoid the problem.

WARNING: A WARNING indicates a potential for property damage, personal injury, or death.

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